Derivatives Sketchbook

Updated November 1, 2026

A beginner-first print companion: explanations, vocabulary, formulas, worked examples, checks, and the practice relevant to this subject.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Your reading path

1. The four derivatives and their payoffs
2. How often interest is added
3. Forward price by no-arbitrage
4. Futures, margin and daily settlement
5. Hedging with futures
6. Forward rates, duration and convexity
7. Swaps: trading payment streams
8. Option strategies and put-call parity
9. Binomial trees for options
10. Black-Scholes and the Greeks

Then: paper labs where available, lesson-check answers, and subject practice.

Lesson 1: The four derivatives and their payoffs

A derivative is a contract whose value comes from something else (a stock, a rate, oil). There are four basic kinds: forwards, futures, swaps and options.

Start with an everyday idea

Imagine agreeing today to buy a bicycle next month for $100. A forward is that commitment. An option is different: you pay for the right to choose later whether to buy.

Words you need

Underlying means the thing the contract depends on. Long means you bought the contract; short means you sold it. A call is a right to buy; a put is a right to sell. Strike is the agreed option price.

Build the explanation

Forward: agree today to buy (long) or sell (short) at a fixed price K on a later date. Both sides must go through with it. Made privately (over the counter).

Future: the same idea, but traded on an exchange, standardized, and settled every day through margin (Class 7).

Swap: trade one stream of payments for another, e.g. fixed interest for floating (Class 9).

Option: the right but not the obligation. A call is the right to buy at K, a put the right to sell at K. You pay a premium up front for that right.

Payoff is what you get at the end. Profit is payoff minus what you paid. Draw the picture every time; it stops sign mistakes.

Formula and units

Long forward payoff = S_T - K
Short forward payoff = K - S_T
Long call payoff = max(S_T - K, 0)
Long put payoff = max(K - S_T, 0)
Profit = payoff - premium paid

S_T = price of the asset at the end, K = agreed (strike/delivery) price.

Walk through the sketch

1. Long forward: you win $1 for every $1 the price ends above K, and lose the same below.

2. Short forward: the mirror image. Your gain is exactly the other side's loss.

3. Long call: you only use it if the price is above K. Dashed line = profit after paying the premium.

4. Long put: you profit when the price falls below K. Insurance for a stock owner.

5. Four shapes to remember: two straight lines, two "hockey sticks".

Worked example

You buy a call with strike $100 for a $5 premium. At expiry the stock is $120. What is your payoff and profit? What if the stock is $90?

1. At $120: payoff = max(120 - 100, 0) = 20

2. Profit = 20 - 5 = 15

3. At $90: payoff = max(90 - 100, 0) = 0 (you just do not use the option)

4. Profit = 0 - 5 = -5. The most a call buyer can lose is the premium.

At $120: payoff 20, profit 15. At $90: payoff 0, profit -5

Common mistakes

1. Mixing payoff with profit.

2. Saying a forward costs money to enter. A fair forward costs nothing up front.

3. Forgetting the short side of an option can lose a lot.

Try before checking

You own a call with strike $100. The asset ends at $120. Its payoff is...

1. $20

2. $120

3. $0

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_02_Lecture_01_Introduction_to_Derivatives.pdf; Hull 8e, Chapters 1 and 10

Lesson 2: How often interest is added

"10% a year" can mean different amounts of money depending on how often interest is added. Continuous compounding is the limit where it is added every instant.

Start with an everyday idea

Interest can earn interest. At 10% a year, $100 becomes $110 after one year and $121 after two, because the second year also earns interest on the first $10.

Words you need

Compounding means adding interest to the balance. Annual means once per year. Continuous compounding is a mathematical limit, not a bank paying you every instant.

Helpful earlier topics: Money today vs money later

Build the explanation

If the bank adds interest twice a year, you earn interest on your interest sooner, so you end up with a bit more. Same headline rate, different result. That is why every rate needs a compounding frequency attached.

m times a year: grow by (1 + R/m) each piece, m pieces a year.

Continuous: grow by eRt. Derivatives math (forwards, Black-Scholes) almost always uses this.

Converting: two rates are "the same" if they grow $1 to the same amount in one year.

Formula and units

Compounded m times: FV = A × (1 + R/m)^(m·t)
Continuous: FV = A × e^(R·t)
Convert m → cont: Rc = m × ln(1 + Rm/m)
Convert cont → m: Rm = m × (e^(Rc/m) - 1)

A = amount, R = annual rate, m = times per year, t = years, ln = natural log.

Walk through the sketch

1. $100 at 10% a year. Added once a year, it jumps to $110 at the end.
2. Added twice a year: $105 after 6 months, then 5% on $105 gives $110.25.
3. Monthly: 12 tiny jumps. You end at $110.47.
4. Continuous: a smooth curve, interest every instant. You end at $110.52.

5. The more often you compound, the more you get, but it levels off at e^0.10.

Worked example

You have 10% a year compounded twice a year. What is the continuously compounded rate?

1. Each half year you grow by 1 + 0.10/2 = 1.05

2. Use the convert rule: Rc = 2 × ln(1.05)

3. ln(1.05) = 0.04879, times 2 = 0.09758

4. Check: e^0.09758 = 1.1025, and 1.05^2 = 1.1025. Same growth.

Rc = 9.758% continuous

Common mistakes

1. Plugging a semiannual rate into e^(rT) without converting it.

2. Using t in months when the rate is per year.

3. Thinking a higher m changes the headline rate. It only changes how much you actually earn.

Try before checking

At 10% annual compounding, $100 becomes what after two years?

1. $120

2. $121

3. $110

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_05_Lecture_04_Interest_Rates.pptx; Hull 8e, Chapter 4 (Interest Rates)

Lesson 3: Forward price by no-arbitrage

The fair forward price is what it costs to buy the asset now with borrowed money and carry it to the delivery date. Any other price gives someone free money.

Start with an everyday idea

If buying an item now and storing it costs $105 by next year, a promise to deliver the same item next year should not cost $120 when you can freely buy, finance and deliver it yourself.

Words you need

Spot is the price now. Forward price is the delivery price agreed today. Arbitrage is a locked-in gain from inconsistent prices under the model’s trading assumptions.

Helpful earlier topics: Money today vs money later, How often interest is added, The four derivatives and their payoffs

Build the explanation

Arbitrage means a riskless profit with no money down. Markets hunt it away. So: if you can borrow $40 at 5%, buy the stock, and hold it a year, it costs you $40 × e^0.05 = $42.05 to own the stock in a year. A forward to buy it in a year must also cost $42.05, otherwise you would do one and the opposite of the other.

Income (dividends) makes holding the stock cheaper, so subtract its PV: F = (S - I)erT.

Dividend yield q or a foreign interest rate rf works the same: F = S e(r-q)T.

Forward price vs forward value: a new forward is worth 0. An old one with delivery price K is worth (F0 - K)e-rT to the long side.

Formula and units

No income: F0 = S0 × e^(rT)
Known income I: F0 = (S0 - I) × e^(rT)
Yield q / FX: F0 = S0 × e^((r - q)T) (q = r_f for currency)
Value of long: f = (F0 - K) × e^(-rT)

S0 = spot price now, r = risk-free rate (continuous), T = years, K = delivery price written in an existing contract.

Walk through the sketch

1. Today the stock costs $40. You can borrow at 5%.
2. Strategy: borrow $40 and buy the stock. Costs you nothing today.
3. In one year you own the stock and owe 40 × e^0.05 = $42.05.

4. If someone offers a forward at $43, sell it to them. Deliver the stock, get 43, repay 42.05. Free $0.95.

5. So the only price with no free money is F0 = $42.05.

Worked example

A stock is $40, pays no dividend, r = 5%, T = 1 year. (a) Fair forward price? (b) The market quotes $43. How do you make money? (c) An old contract has K = $40. What is it worth?

1. (a) F0 = 40 × e^0.05 = 42.05

2. (b) $43 is too high. Sell the forward at 43, borrow $40, buy the stock.

3. In a year: deliver the stock, receive 43, repay the loan 42.05. Profit 0.95 with no risk.

4. (c) f = (42.05 - 40) × e^(-0.05) = 1.95 for the long side.

F0 = 42.05; arbitrage profit 0.95; old contract worth 1.95

Common mistakes

1. Mixing price (the K that makes value 0) and value (what an existing contract is worth).

2. Adding dividends instead of subtracting them.

3. Doing the arbitrage in the wrong direction. Forward too high means sell the forward and buy the asset.

Try before checking

With no income or storage cost, spot $100 and one-year simple interest 5%, the forward price is...

1. $95

2. $100

3. $105

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_03_Lecture_02_Forward_and_Futures_Prices.pptx; FRE6073_SOLUTION_03_Assignment_01_Solutions.pdf

Lesson 4: Futures, margin and daily settlement

With futures, gains and losses are paid every day through a margin account, so nobody can build up a big unpaid loss.

Start with an everyday idea

A futures exchange settles gains and losses each day. Even a sensible hedge can require cash before its final benefit arrives.

Words you need

Margin is collateral, not a down payment on the asset. Daily settlement transfers that day’s gain or loss. A margin call asks you to replenish collateral.

Helpful earlier topics: The four derivatives and their payoffs, Forward price by no-arbitrage

Build the explanation

Initial margin: the deposit to open the position.

Marking to market: each day the exchange moves money between buyers and sellers based on the new futures price.

Maintenance margin: a floor. If your account drops below it you get a margin call and must top it back up to the initial margin.

Clearing house: stands between buyer and seller, so you never depend on the other trader paying. That removes most counterparty risk.

Basis = spot price - futures price. It goes to zero at delivery.

Formula and units

Daily gain (long) = (F_today - F_yesterday) × contract size × number of contracts
If balance < maintenance → deposit (initial - balance)

Short positions get the opposite sign.

Walk through the sketch

1. You open with $12,000. The dashed red line is the $9,000 maintenance floor.
2. Day 1: gold falls $9. You lose $1,800 that day. Balance $10,200.
3. Day 2: down $3 more. Balance $9,600.
4. Day 3: down $5. Balance $8,600 is under the floor. Margin call! Add $3,400 to get back to $12,000.

5. Day 4: gold rises $3. You gain $600. Balance $12,600.

Worked example

You go long 2 gold futures (100 oz each) at $1,250. Initial margin $6,000 per contract, maintenance $4,500. Prices: 1,241 → 1,238 → 1,233 → 1,236. Track the account.

1. Start: 2 × 6,000 = $12,000. Each $1 move = 2 × 100 = $200.

2. Day 1: -9 × 200 = -1,800 → $10,200

3. Day 2: -3 × 200 = -600 → $9,600

4. Day 3: -5 × 200 = -1,000 → $8,600, below the $9,000 floor. Margin call: deposit $3,400 to get back to $12,000.

5. Day 4: +3 × 200 = +600 → $12,600

Margin call of $3,400 on day 3; balance $12,600 after day 4

Common mistakes

1. Topping up only to the maintenance level. You go back to the initial margin.

2. Forgetting to multiply by contract size and number of contracts.

3. Thinking margin is a fee. It is your money, held as a deposit.

Try before checking

You are long one futures contract with multiplier 10. Price falls from 100 to 98. Today’s change is…

1. $20 loss

2. $20 gain

3. $2 loss

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_04_Lecture_03_Futures_Markets_and_Margins.pptx; Hull 8e, Chapter 2

Lesson 5: Hedging with futures

A hedge uses futures to cancel out a price risk you already have. The hedge ratio tells you how many futures to use when the futures and your exposure do not move one-for-one.

Start with an everyday idea

A farmer wants to reduce uncertainty about the selling price of a crop. A futures position can offset some price movements, but the hedge may not match the crop perfectly.

Words you need

A hedge offsets a risk you already have. Basis is spot price minus futures price. Hedge ratio is the amount of protection relative to the exposure.

Helpful earlier topics: Futures, margin and daily settlement, Volatility, diversification, VaR and ES

Build the explanation

Short hedge: you will sell something later (a farmer's wheat). Sell futures now so a price fall is offset by a futures gain.

Long hedge: you will buy something later (an airline's fuel). Buy futures now.

Basis risk: basis = spot - futures. If you hedge with a different but related product (heating oil for jet fuel), or close out early, the basis can move and the hedge is not perfect.

Minimum-variance hedge ratio h*: the slope when you regress changes in your price on changes in the futures price. It is $\rho \times \sigma S/\sigma F$.

Number of contracts: h* × (size of your exposure) / (size of one contract).

Stock index hedge: to remove market risk from a portfolio with beta β, short β × (portfolio value)/(futures price × multiplier) contracts.

Formula and units

$h^* = \rho \times \sigma S / \sigma F$

$N^* = h^* \times QA / QF$

Index hedge: $N^* = \beta \times VA / VF$

σS, σF = std dev of price changes (spot, futures), ρ = their correlation, QA = units exposed, QF = units per contract, VA = portfolio value, VF = futures price × multiplier.

Walk through the sketch

1. Each dot is one month: how much jet fuel moved (up) vs how much heating oil futures moved (across).

2. They move together, but not exactly one-for-one.

3. The best-fit line has slope 0.78. That slope is the hedge ratio h*.

4. So hedge 78% of the 2m gallons: 1.56m gallons ÷ 42,000 = 37 contracts.

Worked example

An airline will buy 2 million gallons of jet fuel. It hedges with heating oil futures (42,000 gallons each). σS = 0.0263, σF = 0.0313, ρ = 0.928. How many contracts?

1. h* = 0.928 × 0.0263 / 0.0313 = 0.7798

2. Exposure to hedge = 0.7798 × 2,000,000 = 1,559,520 gallons

3. Contracts = 1,559,520 / 42,000 = 37.13

4. Round: buy 37 heating oil futures (a long hedge, since it will buy fuel).

h* ≈ 0.78, buy about 37 contracts

Common mistakes

1. Using h = 1 when the futures is on a different product.

2. Going short when you need to buy later (it is a long hedge).

3. Forgetting that a hedge also removes your upside.

Try before checking

You own an asset and fear a price fall. A basic futures hedge is…

1. Sell futures

2. Buy more of the same asset

3. Always buy futures

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_04_Lecture_03_Futures_Markets_and_Margins.pptx; Hull 8e, Chapter 3 (Hedging strategies using futures)

Lesson 6: Forward rates, duration and convexity

Duration tells you how much a bond price drops when rates rise. Convexity is the correction, because the real price curve bends.

Start with an everyday idea

A fixed-rate bond is less attractive when new bonds pay more. Its price tends to fall. Duration estimates how strongly its price responds to a small rate change.

Words you need

Yield is a rate implied by price and payments. Modified duration measures price sensitivity. Convexity adjusts for curvature. A forward rate is a rate implied today for a future period.

Helpful earlier topics: Bonds and building a zero curve

Build the explanation

Forward rate: the rate for a future period that today's zero rates lock in. If 1 year pays 3% and 2 years pays 4%, the second year alone must pay 5% (continuous rates).

Duration is the weighted average time you wait for your cash, where the weights are each payment's share of the price. It is also the sensitivity: a 1% rise in yield cuts the price by about D%.

Convexity: the price-yield curve is bowed. A straight-line (duration) estimate always underestimates the price after a big move either way. Full repricing is the true answer; duration is the quick estimate.

Formula and units

Forward rate: $F(T1,T2) = (R2 \cdot T2 - R1 \cdot T1) / (T2 - T1)$

Duration: $D = \Sigma\ t \times [CF(t) \cdot e^{(-y \cdot t)}] / B$

Price change: $\Delta B \approx -B \times D \times \Delta y\ (+ \frac{1}{2} \times B \times C \times \Delta y^2)$

R = zero rates, y = bond yield (continuous), B = bond price, C = convexity.

Walk through the sketch

1. The price of the bond at each yield. It slopes down and bends (is "convex").
2. At today's yield of 12%, the price is $94.21.
3. Duration draws the tangent line at that point. Slope = -B × D.
4. The tangent is below the curve on both sides. That gap is convexity.
5. For a small move the line is close. For a big move, add the convexity term or reprice fully.

Worked example

3-year bond, 10% coupon paid twice a year, yield 12% continuous (Hull's example). Price, duration, and the effect of a 0.1% rise in yield?

1. Cash flows: $5 at 0.5, 1, 1.5, 2, 2.5 years and $105 at 3 years.

2. Discount each at e^(-0.12t) and add: B = 94.213

3. Weight each time by its share of the price and add: D = 2.653 years

4. ΔB ≈ -94.213 × 2.653 × 0.001 = -0.250, so the price falls to about 93.963.

B = 94.21, D = 2.653, price drops about $0.25

Common mistakes

1. Using Δy = 0.1 instead of 0.001.

2. Forgetting the minus sign: rates up means price down.

3. Using duration for a large rate shock without the convexity term.

Try before checking

Modified duration is 5 and yield rises 0.01 (one percentage point). Approximate price change?

1. -5%

2. +5%

3. -0.05%

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx; Risk syllabus RMN2 (duration, convexity, term structure)

Lesson 7: Swaps: trading payment streams

In an interest rate swap, one side pays fixed and receives floating. You value it as owning one bond and owing another.

Start with an everyday idea

Two borrowers can agree to exchange interest payments: one pays a fixed rate, the other a changing rate. They usually exchange only the net difference in a same-currency interest-rate swap.

Words you need

Notional is the reference amount used to calculate payments. Fixed stays agreed; floating resets to a market rate. A payment stream is a series of payments over time.

Helpful earlier topics: Money today vs money later, Forward rates, duration and convexity

Build the explanation

Company A has a floating-rate loan but wants a fixed payment. It enters a swap: pays a fixed 3% to the bank, receives the floating rate back. Only interest moves; the notional (e.g. $100m) never changes hands in an interest rate swap.

Bond view: receive floating + pay fixed = long a floating bond, short a fixed bond. Value = Bfloat - Bfixed.

Right after a reset, a floating bond is worth its par value (100).

Forward view: treat each future exchange as a forward rate agreement and add them up. Both views give the same answer.

Currency swap: swap interest and principal in two currencies. Value each leg in its own currency, convert at today's spot rate.

Formula and units

Pay fixed, receive floating: $V = B_{float} - B_{fixed}$

$B_{fixed} = \Sigma\, c \cdot e^{(-r_i \cdot t_i)} + L \cdot e^{(-r_n \cdot t_n)}$

$B_{float} = L$ (just after a reset)

L = notional, c = fixed coupon payment, r_i = zero rate for time t_i.

Walk through the sketch

1. Two parties: Company A and a bank.

2. A pays the bank a fixed 3% of $100m every year: $3m.

3. The bank pays A the floating rate (e.g. SOFR) on $100m.

4. Same as A owning a floating bond (worth 100) and owing a fixed bond (worth 99.93).

5. Swap value to A = 100 - 99.93 = +0.07 ($m).

Worked example

$100m swap, 2 years left, you pay 3% fixed annually and receive floating (just reset). Zero rates: 1-yr 2.5%, 2-yr 3.0% (continuous). Value to you?

1. Fixed bond: 3 × e^(-0.025) + 103 × e^(-0.06)

2. = 2.926 + 97.001 = 99.928 ($m)

3. Floating bond right after reset = 100

4. Value = 100 - 99.928 = +0.072 ($m). Positive: the 3% you pay is a little below the fair swap rate for this curve (about 3.04%), so the swap is slightly in your favour.

Swap worth about +$72,000 to the fixed payer

Common mistakes

1. Getting the sign backwards. Write which side you are on first.

2. Forgetting the notional at the end of the fixed bond (103, not 3).

3. For currency swaps, converting with forward rates in the bond view. Use spot.

Try before checking

On $1,000 for one year, you pay 5% fixed and receive 6% floating. Net cash flow?

1. $10 received

2. $10 paid

3. $60 received

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx; FRE6073_ASSIGNMENT_05_Assignment_03.pdf

Lesson 8: Option strategies and put-call parity

Options can be stacked like Lego to make any payoff shape you want. Put-call parity is the rule that links a call and a put with the same strike.

Start with an everyday idea

Combining options is like combining pieces of a safety net: one part limits losses, another can cap gains. Start by adding the payoff of every piece at each possible final price.

Words you need

Premium is the option’s purchase price. A spread combines options. Put-call parity relates matching European calls, puts, stock and a risk-free payment under stated assumptions.

Helpful earlier topics: The four derivatives and their payoffs, Forward price by no-arbitrage

Build the explanation

Bull call spread: buy a low-strike call, sell a high-strike call. Cheaper than a call alone, but profit is capped. You win if the price rises.

Bear spread: the opposite, for a falling price.

Straddle: buy a call and a put at the same strike. You win if the price moves a lot either way. A bet on volatility.

Strangle: like a straddle with two different strikes. Cheaper, needs a bigger move.

Butterfly: buy calls at K1 and K3, sell two at K2. Wins if the price stays near K2 (Class "Volatility smiles" uses it to find probabilities).

Covered call / protective put: stock + short call, or stock + long put (insurance).

Put-call parity: a call plus cash K·e-rT pays the same as a put plus the stock, so they must cost the same. Only for European options.

Bounds: a call is worth at least S - Ke-rT and at most S.

Formula and units

Put-call parity: $c + K \cdot e^{(-rT)} = p + S0$

Call bounds: $\max(S0 - K \cdot e^{(-rT)}, 0) \le c \le S0$

Put bounds: $\max(K \cdot e^{(-rT)} - S0, 0) \le p \le K \cdot e^{(-rT)}$

Profit of a strategy = sum of the payoffs - net premium paid.

Walk through the sketch

1. Start: buy a $30 call. Hockey stick that starts at 30.

2. Add: sell a $35 call. That flips the slope back to flat above 35.

3. Together: the bull spread. Profit is capped at 3, loss is capped at 2.

4. Different idea: a straddle (call + put at the same strike) is a V. It wins on big moves either way.

Worked example

(a) Buy a $30 call for $3 and sell a $35 call for $1. Max profit and max loss? (b) A European call with K = 30, T = 3 months costs $3. S = $31, r = 10%. What must the put cost?

1. (a) Net cost = 3 - 1 = 2. Max loss = 2 (price ends below 30).

2. Max payoff = 35 - 30 = 5, so max profit = 5 - 2 = 3 (price ends above 35).

3. (b) p = c + K·e^(-rT) - S = 3 + 30·e^(-0.025) - 31

4. = 3 + 29.259 - 31 = 1.26

Spread: max profit 3, max loss 2. Put = $1.26

Common mistakes

1. Using put-call parity for American options (it only gives bounds there).

2. Forgetting to subtract the premium when drawing profit.

3. Mixing up which leg is bought and which is sold in a spread.

Try before checking

An option payoff is $12 and you paid $5. Profit, ignoring financing?

1. $12

2. $7

3. $17

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Hull 8e, Chapters 10-11 (Properties of options, trading strategies); FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf

Lesson 9: Binomial trees for options

Let the stock only go up or down each step. Then you can copy the option exactly with stock and cash, and that copy's cost is the option price.

Start with an everyday idea

Replace a complicated future with two possible next prices: up or down. Calculate the option payoff in each state, then work backward to find today’s model price.

Words you need

A node is a point in the tree. Replication means matching cash flows using other assets. Risk-neutral probability is a pricing weight, not a prediction of real-world odds.

Helpful earlier topics: Money today vs money later, The four derivatives and their payoffs

Build the explanation

Replication: hold Δ shares and borrow some cash so the portfolio pays the same as the option in both the up and down state. Δ = (fu - fd)/(Su - Sd).

Risk-neutral probability p: the made-up probability that makes the stock earn the risk-free rate. It is a pricing tool, not anyone's real forecast.

Backward induction: start at the end with payoffs, step back one node at a time: value = e-rΔt[p·fu + (1-p)·fd].

American options: at each node, value = max(hold value, exercise now). Matters most for puts.

With many small steps the tree converges to Black-Scholes (Class 17). Typical choice: u = eσ√Δt, d = 1/u.

Formula and units

p = (e^(rΔt) - d) / (u - d)
f = e^(-rΔt) × [p·f_u + (1 - p)·f_d]
Δ = (f_u - f_d) / (S·u - S·d)
American: f = max(f, exercise value)

u, d = up and down factors, Δt = length of one step, f = option value at a node.

Walk through the sketch

1. Today the stock is $20.

2. After 3 months it is either $22 (up 10%) or $18 (down 10%).

3. After 6 months: $24.20, $19.80 or $16.20.

4. At the end, a call with strike $21 pays 3.20, 0, 0.

5. Step back using p = 0.6523 and discount: 2.03 at the up node, then $1.28 today.

Worked example

Stock $20, u = 1.1, d = 0.9, r = 12%, each step is 3 months. Price a European call with strike $21: one step, then two steps.

1. p = (e^(0.12×0.25) - 0.9) / 0.2 = (1.0305 - 0.9)/0.2 = 0.6523

2. One step: up price 22 → payoff 1; down 18 → payoff 0. c = e^(-0.03) × 0.6523 × 1 = 0.633 (Δ = 1/4 = 0.25 shares)

3. Two steps: ends 24.2, 19.8, 16.2 → payoffs 3.2, 0, 0

4. Up node: e^(-0.03) × 0.6523 × 3.2 = 2.0257; down node = 0

5. Today: e^(-0.03) × 0.6523 × 2.0257 = 1.2822

One-step call = $0.633; two-step call = $1.2822

Common mistakes

1. Using the real-world probability instead of p.

2. Discounting by the whole T at each step instead of Δt.

3. For American puts, forgetting to compare with early exercise at every node.

Try before checking

A call has strike $100. Stock can end at $120 or $80. Payoffs?

1. $20 or $0

2. $120 or $80

3. $20 or -$20

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_01_Binomial_trees.pdf; Hull 8e, Chapter 12

Lesson 10: Black-Scholes and the Greeks

Black-Scholes gives a formula for a European option price from five inputs. The Greeks say how the price moves when one input moves.

Start with an everyday idea

Black-Scholes is a pricing recipe for a European option under specific assumptions. The Greeks ask how that recipe’s answer changes when one input changes a little.

Words you need

European means exercise only at expiry. Sigma (σ) is volatility. Delta measures sensitivity to asset price; vega to volatility. The model’s assumptions can fail in real markets.

Helpful earlier topics: Binomial trees for options

Build the explanation

Idea behind it: hold the option and short Δ shares. Over a tiny instant the randomness cancels, so the hedge must earn the risk-free rate. That gives a PDE whose answer is the formula. Prof. Shimko's "Simpler Option Pricing" lecture reaches the same formula with only college calculus.

Inputs: stock S, strike K, rate r, volatility σ, time T. Note: the expected return μ is not an input.

Delta N(d1): change in option price per $1 move in the stock. Also the hedge ratio.

Gamma: how fast delta changes. High near the strike close to expiry.

Theta: value lost per day as time passes. Vega: change per 1% more volatility. Rho: change per 1% higher rate.

Put-call parity: c + Ke-rT = p + S. Use it to check your answer.

Formula and units

$$c = S \cdot N(d1) - K \cdot e^{(-rT)} \cdot N(d2)$$

$$p = K \cdot e^{(-rT)} \cdot N(-d2) - S \cdot N(-d1)$$

$$d1 = [\ln(S/K) + (r + \sigma^2/2) \cdot T] / (\sigma\sqrt{T})$$

$$d2 = d1 - \sigma\sqrt{T}$$

$$\text{Delta(call)} = N(d1) \quad \text{Gamma} = \varphi(d1)/(S \cdot \sigma \cdot \sqrt{T}) \quad \text{Vega} = S \cdot \varphi(d1) \cdot \sqrt{T}$$

N = cumulative normal (NORM.S.DIST(x,TRUE) in Excel), φ = normal density.

Walk through the sketch

1. At expiry a call with strike $40 is worth max(S - 40, 0): a hockey stick.

2. Before expiry the price is a smooth curve above the stick. The gap is time value.

3. At S = $42 with 6 months left, Black-Scholes says $4.76.

4. The slope of the curve there is delta = 0.78. Move the stock $1, the call moves about $0.78.

5. With 3 months left (dashed) the curve sits lower and hugs the stick. That loss over time is theta. How sharply the curve bends is gamma.

Worked example

S = 42, K = 40, r = 10%, σ = 20%, T = 0.5 years (Hull's example). Find the call, the put and delta.

1. $\sigma\sqrt{T} = 0.2 \times 0.7071 = 0.1414$

2. $d1 = [\ln(1.05) + (0.10 + 0.02) \times 0.5] / 0.1414 = (0.0488 + 0.06)/0.1414 = 0.7693$

3. $d2 = 0.7693 - 0.1414 = 0.6278$; $N(d1) = 0.7791$, $N(d2) = 0.7349$

4. $c = 42 \times 0.7791 - 40 \times e^{(-0.05)} \times 0.7349 = 4.76$

5. Put by parity: $p = 4.76 - 42 + 40e^{(-0.05)} = 0.81$. Delta = 0.779

Call $4.76, put $0.81, delta 0.78

Common mistakes

1. Using σ^2 where σ goes, or forgetting $\sqrt{T}$.

2. Using a non-continuous rate in $e^{(-rT)}$.

3. Thinking the stock's expected return changes the price. It does not.

Try before checking

Delta is 0.6. For a small $1 stock-price rise, the option changes approximately…

1. $0.60

2. $1.60

3. $6

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_12_Simpler_Option_Pricing_Update_2026.pdf; FRE6103_LECTURE_09_Lecture_9_BSM_and_Implied_Densities.pdf; Hull 8e, Chapters 14 and 18

Paper lab: Move money through time

A promised $121 arrives in two years. The annual rate is 10%. Move it backward one year at a time.



Predict and try

1. Predict: should today’s value be greater or smaller than $121?

2. Compute PV for rates 0%, 5% and 10%. Keep the two-year date fixed.

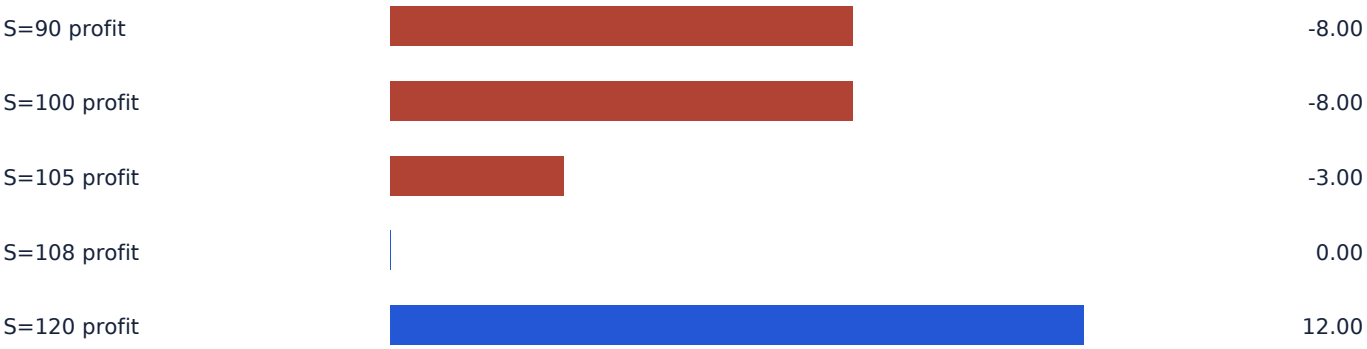
Compare with the explanation

PV = 121/(1+r)^2: $121.00 at 0%, $109.75 at 5%, $100.00 at 10%. A higher discount rate lowers PV. Check: $100 × 1.1 × 1.1 = $121.

Model assumptions: Known future payment; annual compounding; same rate for both years.

Paper lab: Keep option payoff and profit separate

Buy a call on one share with strike $100 and premium $8. Ignore financing of the premium.



Predict and try

1. At a final stock price of $105, would you exercise? Would the whole trade make money?

2. Change the premium to $12. Find the new break-even stock price.

Compare with the explanation

At $105 exercise gives a $5 payoff, but profit = $5 - $8 = -$3. Exercising is better than letting the call expire; it does not recover the premium. Break-even is strike + premium: $108 initially and $112 after the change.

Model assumptions: European call held to expiry; one share; no fees, taxes or interest on the premium.

Paper lab: Rebuild a forward from stock and debt

Buy a $100 share using a loan at 5% continuously compounded. It pays $2 in six months. Deliver the share after one year.



Predict and try

1. Predict: does a dividend raise or lower the fair delivery price?

2. Grow the loan to six months, subtract the dividend, then grow the remaining debt to one year.

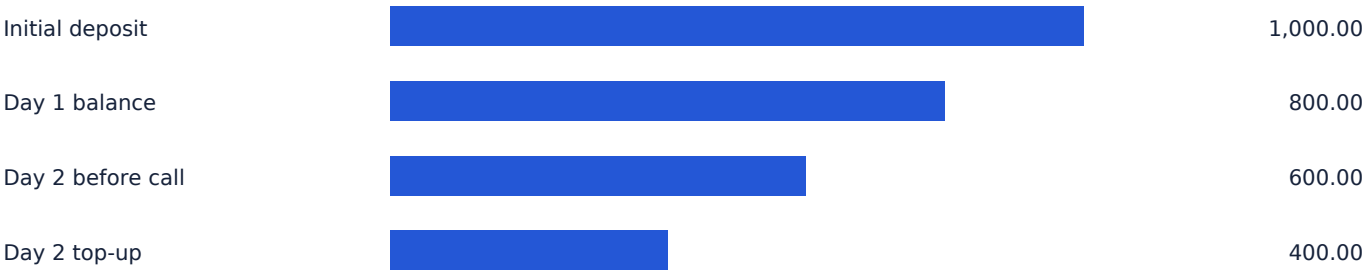
Compare with the explanation

Debt after dividend = 100 exp(0.025) - 2 = $100.5315. At delivery it is 100.5315 exp(0.025) = $103.0765. Without the dividend it would be $105.1271. The dividend lowers the debt you must recover.

Model assumptions: Known dividend; equal borrowing/lending rates; frictionless replication. A forward delivery price is not the current value of an existing forward.

Paper lab: Keep a daily futures ledger

Go long 100 units at quote 100. Deposit $1,000 initial margin. Maintenance is $750. Settlement quotes are 98 and 96.



Predict and try

1. Compute daily trading P/L, then check the maintenance threshold.

2. After restoring initial margin, did the $400 trading loss disappear? Repeat for a short position.

Compare with the explanation

Long: daily P/L is -$200 then -$200. Day 1 balance $800 needs no top-up. Day 2 balance $600 triggers a $400 deposit to restore $1,000. Total deposits $1,400 plus cumulative loss -$400 = balance $1,000. Short: daily gains $200 and $200, final balance $1,400, no calls.

Model assumptions: No interest, fees or withdrawals. A call occurs strictly below maintenance and restores initial margin.

Lesson-check answers

1. The four derivatives and their payoffs

Answer: $20. You may buy for $100 something worth $120: a $20 payoff. Profit also subtracts what you paid for the option.

2. How often interest is added

Answer: $121. Year 1: 100 × 1.10 = 110. Year 2: 110 × 1.10 = 121.

3. Forward price by no-arbitrage

Answer: $105. Borrow $100, buy the asset, and repay $105 in a year. That replication gives a $105 delivery price.

4. Futures, margin and daily settlement

Answer: $20 loss. A long position loses when price falls. (98 - 100) × 10 = -$20.

5. Hedging with futures

Answer: Sell futures. A short futures position tends to gain as futures prices fall, offsetting losses on the asset. Basis risk may remain.

6. Forward rates, duration and convexity

Answer: -5%. Approximate proportional change = -5 × 0.01 = -0.05, or -5%. This is an estimate, not exact repricing.

7. Swaps: trading payment streams

Answer: $10 received. Receive $60, pay $50: net receive $10. The notional itself is not the net payment.

8. Option strategies and put-call parity

Answer: $7. Profit = payoff - premium = 12 - 5 = $7. Payoff alone ignores the purchase cost.

9. Binomial trees for options

Answer: $20 or $0. A call can be left unused. Its payoff is max(stock price - strike, 0).

10. Black-Scholes and the Greeks

Answer: $0.60. Delta × stock change = 0.6 × $1 = $0.60, holding other inputs fixed. Delta itself can change.

Derivatives midterm-style practice

5 questions | 40 points | Learn at your own pace.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Forward price with growing quarterly dividends (8 points)

2. Recover spot from a futures loss (8 points)

3. Find the no-margin-call boundary (8 points)

4. Value an existing bond forward (8 points)

5. Find the fair fixed rate on a swap (8 points)

Question 1: Forward price with growing quarterly dividends

8 points | October 17 paper Q1 · Equity forwards and no-arbitrage

From your paper: stock price $200, continuous risk-free rate 4%, delivery in 3 years. Quarterly dividends begin in 3 months at $1.50; each is 1% above the previous one. Find the fair forward delivery price. State how you treat the dividend at the delivery date.

Show your setup, units, calculation and interpretation.

Question 2: Recover spot from a futures loss

8 points | October 17 paper Q2 · Index futures and dividend yield

From your paper: a long one-year index futures contract has notional 200 × F. Initial index 1,100, continuous dividend yield 4%, risk-free rate 2%. Three months later the investor has lost $100, with no interest on the margin account. Find the index level then.

Show your setup, units, calculation and interpretation.

Question 3: Find the no-margin-call boundary

8 points | October 17 paper Q3 · Daily settlement and collateral

From your paper: short 20 futures contracts, each for 250 index units, initial delivery quote 1,500. Initial margin is 5% of notional; maintenance is 90% of initial margin. Margin earns 4% continuously. Day 1 settlement is 1,498; day 2 is X. Find the largest X with no margin top-up on day 2. State a day-count and interest-timing assumption.

Show your setup, units, calculation and interpretation.

Question 4: Value an existing bond forward

8 points | October 17 paper Q4 · Known income and rate conventions

From your paper: a 10-year bond is worth $920 today. You hold a long forward delivering in one year at $940. The bond pays $80 every six months, including a payment in six months and another just before delivery. Six-month and one-year spot rates are respectively 7% and 8%, quoted annually with semiannual compounding. Find the current contract value.

Show your setup, units, calculation and interpretation.

Question 5: Find the fair fixed rate on a swap

8 points | October 17 paper Q5 · Term structure and swaps

From your paper: annual-compounded spot rates for years 1-6 are 7.0%, 7.3%, 7.7%, 8.1%, 8.4% and 8.8%. A six-year, $10 million plain-vanilla swap pays annually; each floating rate is fixed at the start of its year. Find the fixed annual rate that makes the two legs equal.

Show your setup, units, calculation and interpretation.

Solution 1: Forward price with growing quarterly dividends

First, the words

A forward price is the delivery price that gives a new contract zero value. Known dividends belong to the stock holder before delivery, so remove their present value before growing the stock’s remaining value to delivery.

A smaller problem first

If a stock costs $100 and pays a known $2 dividend before delivery, the forward buyer will not receive that dividend. Subtract its value today first. With continuous interest, moving money forward uses exp(rt); moving it backward uses exp(-rt). A quarter is 0.25 years.

Hint

List twelve dates: 0.25, 0.50, …, 3.00 years. Dividend k is 1.50 × 1.01^(k-1). Discount each dividend separately; then apply the forward-price formula.

Work through it

1. Assume all 12 dividends, including the one at year 3, are paid before delivery. If delivery precedes the final dividend, exclude that payment and explain the different convention.

2. For k = 1,...,12, set tk = k/4 and Dk = 1.50 × 1.01^(k-1). The first dividend is $1.50, not $1.515.

3. Dividend present value I = Σ Dk exp(-0.04tk) = $17.81603. This is a finite sum; the dividends grow but their dates still need separate discounting.

4. Forward price F0 = (S0 - I) exp(rT) = (200 - 17.81603) exp(0.04 × 3) = $205.41185, about $205.41.

5. Check: without dividends the forward would be $225.50. Dividends lower the delivery price. If the year-3 dividend is excluded, add its terminal amount, approximately $1.67, to the result.

6. Transfer drill: if the dividend pattern stays fixed but S0 rises $10, F0 rises 10 × exp(0.12), about $11.27. Explain this without recalculating all dividends.

Mark your reasoning

1. 2: dates, dividend growth and timing assumption

2. 2: discounted dividend sum

3. 3: no-arbitrage formula and $205.41

4. 1: economic direction check

Syllabus: equity forwards, arbitrage and fair-value derivation. Hull 8e Chapter 5: known-income assets. Source photo page 1. Solution is independently derived.

My mark: ____ / 8. My next correction: _____

Solution 2: Recover spot from a futures loss

First, the words

A contract multiplier converts a one-point futures move into a dollar gain or loss. Spot is the index now; futures is its delivery-price quote. Dividend yield q is a continuous proportional payout rate, unlike a known dollar dividend.

A smaller problem first

For a long contract with multiplier 200, a 0.5-point futures fall loses $100: 200 × (-0.5). After three months, a one-year contract has nine months, or 0.75 years, remaining.

Hint

Find the original futures quote using F0 = S0 exp((r-q)T). Use the loss and multiplier to find the new futures quote. Only then convert it back to spot using remaining maturity.

Work through it

1. Assume one contract, multiplier 200 dollars per index point, unchanged rates/yield and the usual deterministic-rate spot/futures relation.

2. Original futures quote F0 = 1,100 exp((0.02 - 0.04) × 1) = 1,078.21854. A yield greater than financing cost puts this quote below spot.

3. Cumulative settlement loss: 200(Ft - F0) = -100. Therefore Ft = F0 - 0.5 = 1,077.71854.

4. Remaining maturity is 1 - 0.25 = 0.75 years. Ft = St exp((0.02 - 0.04) × 0.75).

5. St = 1,077.71854 exp(0.015) = 1,094.00617, about 1,094.01. Do not subtract 100 directly from the index.

6. Transfer drill: for a short position with a $100 loss, the futures quote instead rises 0.5 points. Explain both the sign change and the remaining-time conversion.

Mark your reasoning

1. 2: original quote

2. 2: multiplier and sign of loss

3. 2: remaining maturity and inversion

4. 2: 1,094.01 with stated assumptions

Syllabus: futures mechanics, uses and equity fair-value pricing. Hull 8e Chapters 2 and 5. Source photo page 1. Margin interest is explicitly zero in this question.

My mark: ____ / 8. My next correction: _____

Solution 3: Find the no-margin-call boundary

First, the words

Initial margin is the opening collateral. Maintenance margin is the threshold below which more cash must be added. A short futures position gains when the settlement quote falls. Interest earned on collateral is a separate cash flow.

A smaller problem first

If you are short 2 contracts, each worth $10 per point, a 3-point fall gains 2 × 10 × 3 = $60. A later rise reverses part of that gain. Keep a running balance rather than treating each day in isolation.

Hint

The total multiplier is 20 × 250 = 5,000. Begin with margin, add a day of interest and day 1’s short-position gain. Carry that balance to day 2 and solve an inequality against maintenance margin.

Work through it

1. Assume one day is 1/365 year, interest accrues on the previous end-of-day balance before each daily settlement, and the stated index values are the settlement quotes used for marking to market. The paper does not specify the day-count convention.

2. Notional = 20 × 250 × 1,500 = $7,500,000. Initial margin M0 = 5% × 7,500,000 = $375,000. Maintenance = 90% × 375,000 = $337,500.

3. Day 1 gain for the short = 5,000 × (1,500 - 1,498) = $10,000. M1 = 375,000 exp(0.04/365) + 10,000 = $385,041.10.

4. Day 2 balance M2 = M1 exp(0.04/365) + 5,000(1,498 - X). No call requires M2 ≥ 337,500.

5. Solve at equality: Xmax = 1,498 + [M1 exp(0.04/365) - 337,500]/5,000 = 1,507.51666, about 1,507.52.

6. Check: an increase above Xmax lowers a short’s margin balance below the threshold. If quotes must be on a specified tick grid, round downward to the largest allowed quote.

7. Transfer drill: without margin interest, the boundary is 1,507.50. With a different stated day fraction δ, replace both 1/365 terms by δ. Do not hide the convention.

Mark your reasoning

1. 2: initial and maintenance margin

2. 2: daily short-position gains/losses

3. 2: interest-aware inequality

4. 2: boundary and explicit conventions

Syllabus: futures market mechanics and fair-value applications. Hull 8e Chapter 2; supplied margin lecture. Source photo page 1. Different day-count assumptions produce slightly different numerical boundaries.

My mark: ____ / 8. My next correction: _____

Solution 4: Value an existing bond forward

First, the words

The old delivery price K stays fixed even when the fair price for a new contract changes. Contract value is what the existing position is worth today. A nominal annual rate compounded semiannually is divided by 2 for each half-year period.

A smaller problem first

If the fair new delivery price is $95 but your long contract forces you to pay $100, your contract is worth a negative amount. Discount the $5 disadvantage back to today. Coupon income before delivery belongs to the bond holder, not the forward buyer.

Hint

Subtract the present values of the two coupons from the bond price, then subtract the discounted delivery payment. Use the six-month rate for the first coupon and the one-year rate for the year-one cash flows.

Work through it

1. Assume “own a forward” means a long position and $920 is the cash/dirty bond price used for valuation. If it were a clean quote, accrued interest would be required; it is not provided.

2. Six-month discount factor = $1/(1 + 0.07/2) = 1/1.035$. First coupon PV = $80/1.035 = \$77.29469$.

3. One-year discount factor = $1/(1 + 0.08/2)^2 = 1/1.04^2$. Second coupon PV = $73.96450; delivery-payment PV = $940/1.04^2 = \$869.08284$.

4. Long forward value f0 = 920 - 77.29469 - 73.96450 - 869.08284 = -$100.34202, about -$100.34.

5. Cross-check: the fair new forward price is $(920 - 77.29469 - 73.96450) \times 1.04^2 = \$831.47007$. Then f0 = $(831.47007 - 940)/1.04^2$.

6. Transfer drill: if K falls $10 with everything else unchanged, the long’s value rises $10/1.04^2$, about $9.25. The bond’s 10-year maturity does not make the forward a 10-year contract.

Mark your reasoning

1. 2: two correct discount factors

2. 2: remove both pre-delivery coupons

3. 3: current long value -$100.34

4. 1: distinguish contract value from new forward price

Syllabus: equity/interest-rate forwards, rate conventions and arbitrage pricing. Hull 8e Chapters 4-5. Source photo page 1. The final coupon is explicitly before delivery.

My mark: ____ / 8. My next correction: _____

Solution 5: Find the fair fixed rate on a swap

First, the words

A par swap starts with equal fixed-leg and floating-leg values. Discount factors value $1 at different dates. The par rate is a weighted rate across the payment dates, not the last spot rate or an arithmetic average.

A smaller problem first

For a one-year swap with one payment and discount factor D1, K = (1 - D1)/D1. If D1 = 1/1.05, K is 5%. With several payment dates, add their discount factors in the denominator.

Hint

For each year t calculate Dt = (1 + st)^(-t). With annual accrual and the standard single-curve par-swap assumptions, K ΣDt = 1 - D6.

Work through it

1. Assume a new swap at a reset date, annual accrual factors equal to 1, a common curve for discounting and forwarding, and no credit/funding adjustment.

2. Discount factors D1…D6 are approximately 0.934579, 0.868561, 0.800484, 0.732314, 0.668119 and 0.602874.

3. Under these assumptions, floating coupon-leg PV per dollar of notional is 1 - D6. Fixed coupon-leg PV is K(D1 + … + D6).

4. Equate the legs: K = (1 - 0.6028741017)/(0.9345794393 + 0.8685614624 + 0.8004844320 + 0.7323138101 + 0.6681186409 + 0.6028741017).

5. K = 0.08620181676, or about 8.62018% annually. Fixed yearly cash payment is approximately $862,018.17 on $10 million.

6. Check: notional cancels from the fair-rate equation. Doubling notional doubles payments but leaves K unchanged.

7. Transfer drill: with semiannual payments, use half-year accrual factors and a curve at those payment dates. Do not simply halve the annual par rate without repricing.

Mark your reasoning

1. 2: rate convention and discount factors

2. 2: correct fixed/floating PV equation

3. 3: 8.62018% and payment interpretation

4. 1: assumptions and notional cancellation

Syllabus: swaps, applications and theoretical fair value. Hull 8e Chapters 4 and 7. Source photo page 2. This is the classroom single-curve framework.

My mark: ____ / 8. My next correction: _____

Derivatives mock final

8 questions | 100 points | 120 minutes suggested practice time.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Price a forward and test an arbitrage (12 points)
2. Separate margin deposits from trading profit (12 points)
3. Set and revalue a plain interest-rate swap (12 points)
4. Use parity and distinguish payoff from profit (12 points)
5. Work backward through an American option tree (12 points)
6. Connect a delta hedge to Black-Scholes (14 points)
7. Scale Brownian shocks and interpret log returns (14 points)
8. Explain what a delta hedge leaves behind (12 points)

Question 1: Price a forward and test an arbitrage

12 points | Syllabus: forward fair value and arbitrage; extends midterm Q1.

A stock is $100. It pays a known $2 dividend in six months. The continuously compounded risk-free rate is 5% and delivery is in one year. Find the fair forward price. A dealer instead quotes $106. Describe the cash-and-carry trade and its terminal profit per share. Assume borrowing/lending at 5%, no fees, and the dividend is certain.

Show your setup, units, calculation and interpretation.

Question 2: Separate margin deposits from trading profit

12 points | Syllabus: futures mechanics; extends midterm Q2-Q3.

You short two futures contracts at 4,000. Each has a $50 multiplier per index point. Total initial margin is $2,000 and total maintenance margin is $1,500. Settlement quotes are 4,003 on day 1 and 4,006 on day 2. Margin earns no interest. Show daily P/L, balances before deposits, any required deposit, cumulative P/L and total deposits. Calls occur strictly below maintenance.

Show your setup, units, calculation and interpretation.

Question 3: Set and revalue a plain interest-rate swap

12 points | Syllabus: swap applications and fair value; extends midterm Q5.

A three-year swap has annual payments, $1m notional and discount factors D1 = 0.96, D2 = 0.91, D3 = 0.85. Calculate the par fixed rate K. In a hypothetical immediate repricing before any floating coupon has fixed, the factors become 0.95, 0.88, 0.80. Find the value to the fixed-rate payer. Assume the floating leg resets at valuation, a single projection/discount curve, and unit accrual periods.

Show your setup, units, calculation and interpretation.

Question 4: Use parity and distinguish payoff from profit

12 points | Syllabus: option basics and strategies.

A non-dividend-paying stock is $100. A European call with strike $100 costs $9 and the strike discount factor is 0.95. Find the matching put price and explain put-call parity as two equal terminal portfolios. Separately, a bull call spread buys a strike-90 call for $15 and sells a strike-110 call for $4. Find its payoff and simple profit at stock prices 80, 100 and 120, plus maximum loss and gain. Ignore financing of premiums for the spread.

Show your setup, units, calculation and interpretation.

Question 5: Work backward through an American option tree

12 points | Syllabus: option modeling and fair value; extension beyond the midterm.

A stock starts at $100. Each of two periods multiplies it by u = 1.2 or d = 0.8. The risk-free growth factor is R = 1.05 each period. Price a strike-$105 American put and the corresponding European put. Show the terminal payoffs, pricing probability, continuation values and early-exercise decision. Assume no dividends and a frictionless tree.

Show your setup, units, calculation and interpretation.

Question 6: Connect a delta hedge to Black-Scholes

14 points | Syllabus: fair-value derivation, Black-Scholes and Greeks.

Assume $dS = \mu S\,dt + \sigma S\,dW$ and a smooth European option $V(S,t)$, no dividends, frictionless continuous hedging and constant r, σ. Using Ito's lemma, explain why a position long V and short V_S shares earns r locally and derive the BSM PDE. Then value a one-year call with $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, given $N(d1) = 0.63683$, $N(d2) = 0.55962$ and $\exp(-0.05) = 0.951229$. State delta and two model limitations.

Show your setup, units, calculation and interpretation.

Question 7: Scale Brownian shocks and interpret log returns

14 points | Syllabus: Wiener processes underlying option models.

For geometric Brownian motion with S0 = 100, μ = 6% annually and σ = 20% annually, use Ito to give the distribution of ln(S_T/S0). At T = 0.25 years, find its mean and standard deviation. Calculate S_T for a standard-normal draw Z = -1. Explain which drift changes under risk-neutral valuation if r = 4% and there are no dividends.

Show your setup, units, calculation and interpretation.

Question 8: Explain what a delta hedge leaves behind

12 points | Syllabus: option Greeks and derivative uses.

A dealer is short 100 calls, each covering 100 shares. Each per-share call has delta 0.60, gamma 0.02 per dollar, and vega $0.15 per one percentage-point volatility increase. How many shares make the position initially delta neutral? Estimate the hedged P/L for a $2 stock rise at unchanged volatility using $0.5\,\Gamma(\Delta S)^2$. Separately estimate P/L for a 3-percentage-point volatility rise at unchanged stock price. Ignore time decay, funding and higher-order terms. Explain the hedge limitations.

Show your setup, units, calculation and interpretation.

Solution 1: Price a forward and test an arbitrage

First, the words

A forward fixes a future purchase price. A known dividend belongs to the stock owner before delivery. Arbitrage means matching cash flows with no net risk under the stated assumptions.

A smaller problem first

Without dividends, buying a $100 stock with a 5% continuously compounded loan creates a one-year cost of 100 exp(0.05) = $105.13.

Hint

Borrow the stock price today; use the dividend to reduce the loan at six months.

Work through it

1. Dividend PV = 2 exp(-0.05 × 0.5) = $1.9506.

2. Fair delivery price = (100 - 1.9506) exp(0.05) = $103.0765.

3. Borrow $100, buy one share, and sell a one-year forward at $106. Net initial cash is zero.

4. At six months use the $2 dividend to reduce the debt. At maturity deliver the share for $106 and repay 100 exp(0.05) - 2 exp(0.025) = $103.0765.

5. Locked terminal profit = $2.9235. This relies on identical timing and frictionless funding.

Mark your reasoning

1. 3: dividend timing and PV

2. 3: fair forward price

3. 4: all trade cash flows

4. 2: profit and assumptions

Hull: forward prices with known income; supplied forward-pricing lecture.

My mark: ____ / 12. My next correction: _____

Solution 2: Separate margin deposits from trading profit

First, the words

Short means you gain when the futures quote falls. Maintenance margin is the trigger; initial margin is the target after a call.

A smaller problem first

A short position of 100 units loses $100 when the quote rises by $1. A deposit of $100 covers collateral but does not undo that loss.

Hint

Daily P/L = -2 × 50 × (new quote - previous quote).

Work through it

1. Total exposure is 100 dollars per point; use the short-position minus sign.

2. Day 1: P/L = -$300; balance = $1,700. No call.

3. Day 2: P/L = -$300; balance before deposit = $1,400, below $1,500.

4. Deposit $600 to restore $2,000. Total deposits, including the original $2,000, are $2,600.

5. Cumulative trading P/L = -$600. Check: deposits $2,600 plus P/L -$600 = final balance $2,000.

Mark your reasoning

1. 2: exposure and sign

2. 3: day 1

3. 3: day 2 and margin call

4. 4: cumulative loss and balance identity

Hull: futures markets and margin; supplied futures-markets lecture.

My mark: ____ / 12. My next correction: _____

Solution 3: Set and revalue a plain interest-rate swap

First, the words

A discount factor is the price today of $1 paid later. A par swap has zero value at inception. A payer swap pays fixed and receives floating.

A smaller problem first

If annual discount factors are 0.96 and 0.90, a two-year par fixed rate is (1 - 0.90)/(0.96 + 0.90).

Hint

At reset, PV of floating coupons is N(1 - D3); PV of fixed coupons is NK times the sum of discount factors.

Work through it

1. Original annuity factor = 0.96 + 0.91 + 0.85 = 2.72.

2. K = (1 - 0.85)/2.72 = 0.05514706, or 5.5147% annually.

3. After repricing, floating coupon PV = 1,000,000 × (1 - 0.80) = $200,000.

4. Fixed coupon PV = 1,000,000 × K × 2.63 = $145,036.76.

5. Payer value = floating minus fixed = +$54,963.24. Away from a reset, account for the already-fixed next floating coupon.

Mark your reasoning

1. 3: original annuity and par rate

2. 3: floating PV

3. 3: fixed PV

4. 3: payer sign and reset assumption

Hull: interest-rate swaps; supplied duration/swaps lecture.

My mark: ____ / 12. My next correction: _____

Solution 4: Use parity and distinguish payoff from profit

First, the words

Payoff is the contract cash at expiry. Profit subtracts its initial cost; this question explicitly ignores financing of that cost.

A smaller problem first

A call with strike $100 has payoff $20 at a stock price of $120. If its premium was $8, simple profit is $12.

Hint

Parity: call + discounted strike = put + stock. Spread payoff is the difference of two call payoffs.

Work through it

1. Put = 9 + 100 × 0.95 - 100 = $4.

2. Call plus a bond paying $100 and stock plus put both pay max(stock price, 100) at expiry; their prices must match.

3. Spread premium paid = 15 - 4 = $11. Payoff = max(S - 90, 0) - max(S - 110, 0).

4. At S = 80, 100, 120: payoffs are $0, $10, $20; simple profits are -$11, -$1, +$9.

5. Maximum loss is $11; maximum gain is $9. Profit and payoff are different graphs.

Mark your reasoning

1. 3: put price

2. 3: replication explanation

3. 4: three payoffs and profits

4. 2: maximum gain/loss

Hull: properties of stock options and trading strategies.

My mark: ____ / 12. My next correction: _____

Solution 5: Work backward through an American option tree

First, the words

Risk-neutral probabilities price tradable claims; they need not be forecasts. An American option allows early exercise.

A smaller problem first

At any American put node, compare the discounted continuation value with immediate exercise max(K - S, 0). Keep the larger.

Hint

q = (R - d)/(u - d). The period-1 down node is the important exercise decision.

Work through it

1. q = (1.05 - 0.8)/(1.2 - 0.8) = 0.625; 0 < q < 1.

2. Terminal stock prices are 144, 96, 64. Put payoffs are 0, 9, 41.

3. At S = 120, continuation = (0.625 × 0 + 0.375 × 9)/1.05 = $3.2143; immediate exercise is 0.

4. At S = 80, continuation = (0.625 × 9 + 0.375 × 41)/1.05 = $20; immediate exercise is $25. American exercise wins.

5. American root value = (0.625 × 3.2143 + 0.375 × 25)/1.05 = $10.8418, greater than immediate exercise $5.

6. European root uses $20 at the down node: value = $9.0561. Early-exercise flexibility is worth $1.7857.

Mark your reasoning

1. 2: q

2. 2: terminal payoffs

3. 4: both continuation/exercise comparisons

4. 2: American root

5. 2: European comparison

Hull: binomial trees and American exercise; supplied Binomial_trees.pdf. This adds depth beyond the short app lesson.

My mark: ____ / 12. My next correction: _____

Solution 6: Connect a delta hedge to Black-Scholes

First, the words

Delta is the local change in option value per $1 stock move. A locally riskless hedge removes the Brownian shock, not all future risk without rebalancing.

A smaller problem first

If a call delta is 0.6, buying 0.6 shares offsets the immediate stock sensitivity of one short call.

Hint

Ito gives $dV = (V_t + \mu S V_S + 0.5\sigma^2S^2V_{SS})dt + \sigma S V_S dW$. Subtract $V_S dS$.

Work through it

1. The self-financing hedged change is $dV - V_S dS = (V_t + 0.5\sigma^2S^2V_{SS})dt$; the dW and μ terms cancel.

2. The hedge value is $V - S V_S$. No arbitrage requires its drift to equal $r(V - S V_S)$.

3. Rearrange: $V_t + 0.5\sigma^2S^2V_{SS} + rS V_S - rV = 0$, with terminal condition $V(S,T) = \max(S - K, 0)$ for a call.

4. $C = S N(d1) - K \exp(-rT) N(d2) = 100 \times 0.63683 - 100 \times 0.951229 \times 0.55962 \approx \$10.45$.

5. Delta = $N(d1) = 0.63683$. For one short option on one share, buy that many shares initially and rebalance.

6. Constant volatility omits smiles and volatility changes; continuous frictionless hedging omits costs, gaps and discrete rebalancing. These limitations matter for actual hedging.

Mark your reasoning

1. 4: cancellation and self-financing reasoning

2. 3: PDE and terminal condition

3. 3: numerical call

4. 2: delta

5. 2: two limitations

Hull: Wiener processes, Ito and the Black-Scholes model. Derivation is an original scaffold, not a reproduced exam answer.

My mark: ____ / 14. My next correction: _____

Solution 7: Scale Brownian shocks and interpret log returns

First, the words

A Brownian increment over length Δt is sqrt(Δt) times a standard-normal draw. Drift scales with time; uncertainty scales with its square root.

A smaller problem first

Annual volatility 20% becomes 10% over a quarter: 20% × sqrt(0.25). It is not 5%.

Hint

For f(S) = ln S, f_S = 1/S and f_SS = -1/S². Include the second-order Ito term.

Work through it

1. Ito gives d ln S = (μ - 0.5σ²)dt + σ dW.

2. Thus ln(S_T/S0) = (μ - 0.5σ²)T + σ sqrt(T) Z, with Z standard normal.

3. Quarter-year mean = (0.06 - 0.5 × 0.20²) × 0.25 = 0.01. Standard deviation = 0.20 sqrt(0.25) = 0.10.

4. For Z = -1, log return = 0.01 - 0.10 = -0.09, so S_T = 100 exp(-0.09) = $91.39.

5. Under the risk-neutral measure replace μ by r = 0.04. Log-return mean becomes 0.005; volatility remains 0.10 in this model.

6. The physical drift describes the assumed real-world process; the pricing drift enforces no-arbitrage valuation. They answer different questions.

Mark your reasoning

1. 4: Ito log transformation

2. 3: distribution and time scaling

3. 3: numerical price

4. 2: risk-neutral drift

5. 2: interpretation

Hull: Wiener processes and Ito; Valuation stochastic-process lecture for supporting intuition.

My mark: ____ / 14. My next correction: _____

Solution 8: Explain what a delta hedge leaves behind

First, the words

Gamma measures how delta changes as the stock moves. Vega measures sensitivity to a volatility change. Here vega is quoted per one percentage point of volatility.

A smaller problem first

A share has delta 1 and gamma 0. Shares can neutralize delta but cannot cancel option gamma.

Hint

There are 10,000 option-share units. Short positions reverse the Greeks' signs.

Work through it

1. Option position delta = -10,000 × 0.60 = -6,000. Buy 6,000 shares.

2. Total option gamma = -10,000 × 0.02 = -200 per dollar. The stock adds zero gamma.

3. After canceling the first-order delta term, estimated stock-move P/L = 0.5 × (-200) × 2^2 = -$400.

4. Total vega = -10,000 × 0.15 = -$1,500 per volatility percentage point. A 3-point increase gives approximately -$4,500.

5. Delta neutrality is local and temporary. Gamma changes delta, volatility changes option prices, and discrete hedging/costs prevent a perfect continuous hedge.

Mark your reasoning

1. 3: multiplier and share hedge

2. 3: gamma P/L

3. 3: vega units and P/L

4. 3: limitations

Hull: Greeks and hedging. Supplied syllabus lists Greeks in the final option block.

My mark: ____ / 12. My next correction: _____

Valuation Sketchbook

Updated November 1, 2026

A beginner-first print companion: explanations, vocabulary, formulas, worked examples, checks, and the practice relevant to this subject.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Your reading path

1. Money today vs money later
2. Loans: payments, default and prepayment
3. Bonds and building a zero curve
4. Capital budgeting: NPV and IRR
5. Simulation, correlation and Cholesky
6. Random walks and stock price paths
7. Portfolio theory and the CAPM
8. ETFs and futures for end users
9. CDO waterfalls and tranches
10. Volatility smiles and implied densities
11. Real options and the Merton model

Then: paper labs where available, lesson-check answers, and subject practice.

Lesson 1: Money today vs money later

A dollar later is worth less than a dollar today, because today's dollar could sit in a bank and grow. Present value (PV) just undoes that growth.

Start with an everyday idea

You can choose $100 today or $105 next year. If your bank pays 5%, those choices have the same value in this simplified example.

Words you need

Present value means today’s equivalent amount. A rate of 5% is 0.05 in a formula. A cash flow is money paid or received.

Build the explanation

Think of the bank as a time machine for money. Put $V0 in at 5% and next year you get $V0 × 1.05. Present value runs the machine backwards: divide instead of multiply.

Future value (FV): what money grows into. Multiply by (1 + r) each year.

Present value (PV): what future money is worth today. Divide by (1 + r) each year.

Discount factor: the number you multiply a future cash flow by to get its PV. For 3 years at 5% it's $1/1.05^3 = 0.8638$.

Annuity: the same payment every period. Perpetuity: the same payment forever, PV = C / r.

Prof. Shimko's point ("PV fails"): PV only works cleanly when you know the cash flows and the right rate. Risky cash flows need a risk adjustment, which later classes cover.

Formula and units

PV = CF / (1 + r)^t

Perpetuity: PV = C / r

Annuity: PV = C × [1 - (1+r)^(-n)] / r

CF = cash flow, r = rate per period, t = number of periods, C = repeating payment.

Walk through the sketch

1. This is a timeline. Today is year 0. Someone promises you $100 at year 3.
2. Move the $100 back one year: divide by 1.05. It shrinks to $95.24.
3. Back one more year: divide by 1.05 again. Now $90.70.

4. Back to today: $86.38. That is the present value. Smaller coin, same "worth".

5. Shortcut: do all three divides at once. PV = 100 ÷ 1.05^3 = 86.38.

Worked example

You will get $100 in 3 years. The bank pays 5% a year. What is it worth today?

1. Write the formula: PV = 100 / 1.05^3

2. Work out 1.05^3 = 1.05 × 1.05 × 1.05 = 1.157625

3. Divide: 100 / 1.157625 = 86.38

4. Check by going forward: 86.38 × 1.157625 = 100. It works.

PV = $86.38

Common mistakes

1. Using an annual rate with monthly cash flows. Match the rate to the period (6% nominal annually with monthly compounding = 0.5% per month).

2. Mixing up multiply (future) and divide (present).

3. Forgetting that a perpetuity's first payment comes one period from now, not today.

Try before checking

What is $110 next year worth today at 10%?

1. $100

2. $110

3. $121

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_01_Lecture_1_Present_Value_Fails.pdf; FRE6103_BOOK_03_Valuation_Principles_Ch_1_2.pdf

Lesson 2: Loans: payments, default and prepayment

A loan payment is split into interest first, then the rest pays down what you owe. A pool of loans also loses money to people who default or pay off early.

Start with an everyday idea

When you repay a loan, part of each payment pays for borrowing and the rest reduces what you owe. Early payments often contain more interest because the outstanding balance is larger.

Words you need

Principal is the amount still owed. Interest is the borrowing charge. Default means failing to pay as agreed. Prepayment means paying earlier than scheduled.

Helpful earlier topics: Money today vs money later, How often interest is added

Build the explanation

This is the lecture's own example: borrow $5,000 for 24 months at 6% a year. Every month you pay the same amount. Early on most of it is interest, later most of it is principal, because the balance (and so the interest) keeps shrinking.

Amortization: paying the loan down to zero with equal payments.

Borrower's view: what does it cost me? Lender's view: what cash do I actually get back?

Default: some borrowers stop paying. In the course model, a fixed share of the remaining pool defaults each month (0.5% in Mini Project 1).

Prepayment: some borrowers pay everything early (1% a month in the project). The lender gets cash sooner but loses future interest.

Formula and units

Payment: PMT = B × i / [1 - (1+i)^(-n)]
Each month: interest = balance × i
principal paid = PMT - interest
new balance = balance - principal paid
Pool left after k months ≈ (1 - d - p)^k

B = amount borrowed, i = monthly rate (6%/12 = 0.5%), n = number of months, d = monthly default share, p = monthly prepay share.

Walk through the sketch

1. You borrow $5,000. That tall bar is what you owe at month 0.
2. Each month you pay $221.60. Red part = interest, blue part = paying down the loan.
3. Over 24 months the balance falls to zero.
4. Notice the red part shrinks and the blue part grows: less balance means less interest.
5. For a lender with a pool of loans, defaults and prepayments make the pool (dashed) shrink faster.

Worked example

Borrow $5,000 for 24 months at 6% a year. What is the payment, and what happens in month 1?

1. Monthly rate i = 6% / 12 = 0.005
2. PMT = 5000 × 0.005 / (1 - 1.005^(-24)) = $221.60 (Excel: PMT(0.06/12,24,-5000))
3. Month 1 interest = 5000 × 0.005 = $25.00
4. Principal paid = 221.60 - 25.00 = $196.60, new balance = $4,803.40

Payment $221.60; balance after month 1 is $4,803.40

Common mistakes

1. Using 6% instead of 0.5% for the month.
2. Forgetting interest is paid before principal.
3. In the pool model, default and prepayment both shrink the balance, but only default loses money.

Try before checking

A $100 balance earns $5 interest and you pay $20. What remains?

1. $85
2. $80
3. $105

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_03_Lecture_2A_Loans_with_exponential_default_risk.pdf; FRE6103_REVIEW_02_Mini_Project_1_Two_Sides_of_a_Loan_Instructions.txt

Lesson 3: Bonds and building a zero curve

A zero-coupon bond pays one amount at one date, so its price tells you today's value of $1 at that date. Bootstrapping peels these discount factors out of coupon bonds one date at a time.

Start with an everyday idea

A bond is an IOU: lend money now, receive promised payments later. Price each payment in today’s money and add them up.

Words you need

A coupon is a scheduled interest payment. Face value is the amount repaid at maturity. A zero-coupon bond has just one payment at the end. A discount factor converts future money into today’s money.

Helpful earlier topics: Money today vs money later, How often interest is added

Build the explanation

A coupon bond is just a bundle of little zero-coupon bonds, one for each payment. If you already know the discount factor for the early dates, you can subtract those pieces from the bond price. What's left must be the value of the last payment, which gives you the next discount factor. Repeat. That is bootstrapping.

Discount factor Z(t): price today of $1 paid at time t.

Zero rate: the rate that matches Z(t). Continuous: r = -ln Z(t) / t.

Zero curve / term structure: zero rates plotted against maturity. Mini Project 2 builds this from Treasury data.

Formula and units

Bond price = Σ CF(t) × Z(t)

Z(t) = e^(-r·t) → r = -ln Z(t) / t

Bootstrap last date: Z(T) = (Price - Σ earlier CF × Z) / CF(T)

CF(t) = cash flow at time t, Z(t) = discount factor, r = zero rate (continuous).

Walk through the sketch

1. Start with the easy one. A 6-month zero costs $98 for $100. So $1 in 6 months is worth $0.98 today.

2. Next, a 1-year coupon bond costing $99. It pays $2 at 6 months and $102 at 1 year.

3. We already know how to value the $2: 2 × 0.98 = $1.96.

4. The rest of the price must be the $102: 99 - 1.96 = 97.04. So Z(1) = 97.04 / 102 = 0.9514.

5. Turn each Z into a rate and plot. That is your zero curve.

Worked example

A 6-month zero sells at $98 per $100. A 1-year bond with a 4% coupon paid twice a year ($2 every 6 months) sells at $99. Find Z(0.5) and Z(1).

1. Zero bond: Z(0.5) = 98 / 100 = 0.98

2. The 1-year bond pays $2 at 0.5 years and $102 at 1 year.

3. Value of the first coupon: 2 × 0.98 = 1.96

4. What is left: 99 - 1.96 = 97.04 = 102 × Z(1), so Z(1) = 0.9514

5. Zero rates: -ln(0.98)/0.5 = 4.04%, -ln(0.9514)/1 = 4.98%

Z(0.5) = 0.98, Z(1) = 0.9514; zero rates 4.04% and 4.98%

Common mistakes

1. Forgetting the final payment includes the coupon (102, not 100).

2. Using the annual coupon (4) instead of the half-year coupon (2).

3. Mixing continuous and annual zero rates.

Try before checking

A zero-coupon bond pays $110 in one year. At 10%, its price is...

1. $100

2. $110

3. $121

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_04_Lecture_2B_Bonds_and_rates.pdf; FRE6103_DATA_08_Treasury_data_090426.xlsx; FRE6073_LECTURE_05_Lecture_04_Interest_Rates.pptx

Lesson 4: Capital budgeting: NPV and IRR

A company should take a project if its NPV is positive: the present value of what it brings in is more than what it costs. IRR is the discount rate where NPV is exactly zero.

Start with an everyday idea

Would you pay $100 now for a project that gives back $110 next year? At a required return of 5%, that future $110 is worth about $104.76 today, so the project adds about $4.76.

Words you need

NPV is present value of benefits minus costs. IRR is a discount rate that makes NPV zero. A required return is the rate used to judge the investment.

Helpful earlier topics: Money today vs money later, Bonds and building a zero curve

Build the explanation

NPV: add up the PV of every cash flow, including the (negative) cost today.

Discount rate: the cost of capital, i.e. the return investors could get elsewhere at the same risk (CAPM from the Portfolio class helps here).

IRR: the rate that makes NPV = 0. If IRR > cost of capital, NPV is positive. Find it by trial and error or Excel's IRR().

When IRR misleads: projects of different size, cash flows that change sign more than once (can have two IRRs), or different timing. When in doubt, trust NPV.

Payback period ignores timing and anything after the cutoff. Use it only as a rough check.

Welch's book also covers capital structure: how mixing debt and equity changes the cost of capital (WACC).

Formula and units

NPV = Σ CF_t / (1 + r)^t (t = 0, 1, 2, ...)

IRR: the r where NPV = 0

WACC = E/V × rE + D/V × rD × (1 - tax)

CF_0 is usually negative (the investment). E, D = market values of equity and debt, V = E + D.

Walk through the sketch

1. Cash flows on a timeline: pay $100 now, get 40, 50, 30 later.

2. Shrink each future amount to today's money at 8%, then add. NPV = +3.72.

3. Now try every discount rate. NPV falls as the rate rises.

4. Where the curve crosses zero (about 10.1%) is the IRR. Our 8% is to the left, so NPV > 0.

Worked example

A project costs $100 today and pays $40, $50 and $30 at the end of years 1, 2 and 3. Cost of capital 8%. NPV? IRR?

1. PV of year 1: 40 / 1.08 = 37.04

2. PV of year 2: 50 / 1.08^2 = 42.87

3. PV of year 3: 30 / 1.08^3 = 23.81

4. NPV = 37.04 + 42.87 + 23.81 - 100 = 3.72 → take it

5. IRR (Excel IRR or trial): about 10.1%, above 8%. Same answer.

NPV = $3.72, IRR ≈ 10.1%

Common mistakes

1. Forgetting the year-0 cost.

2. Choosing between two projects by IRR instead of NPV.

3. Using a company-wide discount rate for a project with very different risk.

Try before checking

Cost $100 today; receive $110 next year; discount rate 10%. NPV?

1. $10

2. $0

3. $110

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf; FRE6103_LECTURE_13_VL10_Multiperiod_cash_flows.pdf

Lesson 5: Simulation, correlation and Cholesky

Simulation means making up thousands of possible futures with random numbers and averaging the results. Cholesky is the trick that makes those random numbers move together.

Start with an everyday idea

Instead of predicting one future, roll many imaginary dice to generate possible futures. Then summarize the outcomes. Related risks need related dice, not independent ones.

Words you need

Simulation generates possible outcomes from assumptions. Correlation describes co-movement. Cholesky is a matrix method for giving random draws a chosen correlation structure.

Helpful earlier topics: Volatility, diversification, VaR and ES

Build the explanation

Some problems are too messy for a formula (10 bonds defaulting at different times, a CDO waterfall). So you let the computer roll dice many times and look at the outcomes.

Uniform U (between 0 and 1) → normal Z with NORM.S.INV(U) in Excel.

Correlation: independent Z's don't move together. To get correlation ρ, mix them: $X2 = \rho \cdot Z1 + \sqrt{(1-\rho^2)} \cdot Z2$. For many variables, the Cholesky matrix does this mixing for all of them at once.

Default time: with a constant default rate λ, a random default time is $\tau = -\ln(U)/\lambda$. If τ is before maturity, that bond defaults in this run.

Sanity checks: the sample mean, standard deviation and correlation should match what you asked for. "Moment matching" forces them to match exactly.

Formula and units

Two variables: $X1 = Z1$

$X2 = \rho \cdot Z1 + \sqrt{(1 - \rho^2)} \cdot Z2$

Many variables: $X = L \cdot Z$ where $L \cdot L^T$ = correlation matrix

Default time: $\tau = -\ln(U) / \lambda$

Z = independent standard normals, L = Cholesky (lower triangular) matrix, λ = default intensity per year.

Walk through the sketch

1. 400 pairs of independent random numbers. A round cloud: knowing one tells you nothing about the other.

2. Apply Cholesky with ρ = 0.6. The cloud stretches into a tilted oval. Now they move together.

3. One dot: (Z1, Z2) = (1.2, -0.5) becomes (1.2, 0.32). It moved up because Z1 was high.

4. Default times: draw U, then τ = -ln(U)/λ. Small U means a long wait; big U means early default.

Worked example

ρ = 0.6, and you drew Z1 = 1.2, Z2 = -0.5. Find X1 and X2. Then: λ = 4% a year and U = 0.7. When does the bond default?

1. X1 = Z1 = 1.2

2. √(1 - 0.36) = 0.8

3. X2 = 0.6 × 1.2 + 0.8 × (-0.5) = 0.72 - 0.40 = 0.32

4. τ = -ln(0.7)/0.04 = 0.3567/0.04 = 8.92 years. For a 5-year bond: no default in this run.

X1 = 1.2, X2 = 0.32; default at 8.92 years

Common mistakes

1. Using a covariance matrix when the formula needs a correlation matrix (or the other way round).

2. Re-drawing random numbers every time Excel recalculates. Fix them (paste values) so results are comparable.

3. Forgetting to check that sample correlation is near the target.

Try before checking

Why correlate simulated stock returns?

1. To guarantee a profit

2. To represent how assets move together

3. To remove uncertainty

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_05_Lecture_4_Simulation.pdf; FRE6103_DATA_02_Lecture_4_Cholesky_five.xlsx; FRE6103_DATA_03_Lecture_4_Simulation_spreadsheets.xlsx

Lesson 6: Random walks and stock price paths

A stock price is modelled as a steady drift plus random shocks. Using percentage shocks (geometric Brownian motion) keeps prices positive and gives a lognormal shape.

Start with an everyday idea

A stock path is like taking many small steps with both a general tendency and surprises. A simulated path is one possible story, not a price forecast you can rely on.

Words you need

Drift is the model’s average direction. Volatility sets the size of surprises. A time step is the interval between simulated prices. Brownian motion is a mathematical model of random changes.

Helpful earlier topics: How often interest is added, Simulation, correlation and Cholesky

Build the explanation

Wiener process (Brownian motion) W: a random walk where each tiny step is normal with variance equal to the time step. Over time Δt the change is ε√Δt.

Arithmetic Brownian motion: dX = μ dt + σ dW. Can go negative. Fine for spreads or rates, not for stock prices.

Geometric Brownian motion (GBM): dS = μS dt + σS dW. Returns are normal, prices are lognormal.

Simulating one step exactly: Snext = S × exp((μ - σ²/2)Δt + σ√Δt·Z). The -σ²/2 is there because of Itô's lemma; without it your average would be too high.

Mean reversion (e.g. rates, commodities): the drift pulls the value back toward a long-run level.

Formula and units

GBM: $dS = \mu \cdot S \cdot dt + \sigma \cdot S \cdot dW$

One step: $S(t+\Delta t) = S(t) \times \exp[(\mu - \sigma^2/2) \cdot \Delta t + \sigma \cdot \sqrt{\Delta t} \cdot Z]$

Expected: $E[S_T] = S0 \times e^{(\mu T)}$

μ = expected return (drift), σ = volatility, Z = standard normal draw.

Walk through the sketch

1. Start at $100. Time runs left to right over one year.
2. One simulated path: small random up and down steps with a slight upward drift.
3. More paths. Each one is a possible future.

4. The blue line is the average path, 100·e^(0.08t). The band shows the typical spread, which widens over time.

5. At year end the prices pile up in a lopsided (lognormal) shape: a long tail upward, never below zero.

Worked example

S = 100, μ = 8%, σ = 20%, Δt = 1 year, and you draw Z = 0.5. What is the simulated price?

1. Drift part: (0.08 - 0.02) × 1 = 0.06

2. Shock part: 0.20 × √1 × 0.5 = 0.10

3. Add: 0.16, then e^0.16 = 1.1735

4. S = 100 × 1.1735 = 117.35

S = $117.35

Common mistakes

1. Forgetting -σ²/2 in the exponent.

2. Using σ × Δt instead of σ × √Δt.

3. Mixing up the median (S0·e^((μ-σ²/2)T)) and the mean (S0·e^(μT)).

Try before checking

Why do two simulated paths differ with the same starting price?

1. They use different random shocks

2. One must contain a mistake

3. The starting price determines the whole future

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_06_Lecture_5_Stochastic_Processes.pdf; FRE6103_DATA_05_Lecture_5_Simulation_spreadsheets.xlsx; Hull 8e, Chapter 13 (Wiener processes and Itô)

Lesson 7: Portfolio theory and the CAPM

Investors only get paid for risk they cannot diversify away. Beta measures that risk, and the CAPM turns beta into a required return.

Start with an everyday idea

Owning an umbrella shop and an ice-cream shop can smooth income because different weather helps each one. Diversification depends on how investments move together.

Words you need

A portfolio is a collection of investments. Beta measures sensitivity to market returns. CAPM links required return to market risk under simplifying assumptions; it is not a guaranteed return.

Helpful earlier topics: Volatility, diversification, VaR and ES

Build the explanation

Efficient frontier: the best possible return for each level of risk, built by mixing stocks (Markowitz). Correlation below 1 bends it to the left.

Capital market line: mix the risk-free asset with the best risky portfolio (the "market" M). It is the line from rf that just touches the frontier. Its slope is the Sharpe ratio.

Beta: how much an asset moves when the market moves. $\beta = \mathrm{Cov}(r_i, r_M)/\mathrm{Var}(r_M)$.

CAPM: $E[r] = r_f + \beta \times (E[r_M] - r_f)$.

Lintner (certainty-equivalent) form: value a risky cash flow X by subtracting a risk charge, then discounting at the risk-free rate. This is how the course values CDO tranches.

Formula and units

Sharpe ratio: $(E[r] - r_f) / \sigma$

CAPM: $E[r_i] = r_f + \beta_i \times (E[r_M] - r_f)$

Lintner: $PV = (E[X] - \lambda \cdot \mathrm{Cov}(X, r_M)) / (1 + r_f)$

$\lambda = (E[r_M] - r_f) / \sigma_M^2$

λ = market price of risk, X = an uncertain cash flow in one period.

Walk through the sketch

1. Each dot is one stock: its risk (σ) and expected return.

2. Mixing them gives the efficient frontier: the best return for each level of risk.

3. Add the risk-free asset at 4%. Draw a line from it that just touches the frontier.

4. The touching point is the market portfolio M. Everyone should hold a mix of M and the risk-free asset.

5. For one asset, CAPM says: required return = 4% + β × 6%.

Worked example

rf = 4%, market return 10%, σ_M = 20%. (a) Required return for β = 1.2? (b) A cash flow has E[X] = $110 and Cov(X, r_M) = 2. Value it with Lintner.

1. (a) 4% + 1.2 × (10% - 4%) = 11.2%

2. (b) λ = 0.06 / 0.04 = 1.5

3. Risk charge = 1.5 × 2 = 3, certainty equivalent = 110 - 3 = 107

4. PV = 107 / 1.04 = 102.88

Required return 11.2%; PV = $102.88

Common mistakes

1. Using total risk σ instead of beta for required return.

2. Discounting the certainty equivalent at a risky rate. It already includes the risk charge, so use rf.

3. Forgetting that correlation, not volatility alone, drives diversification.

Try before checking

Risk-free rate 3%, market premium 5%, beta 2. CAPM required return?

1. 8%

2. 13%

3. 10%

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_07_Lecture_6_MPT_and_the_CAPM.pdf; FRE6103_DATA_06_MPT_Ten_Stock_Portfolio.xlsx; FRE6103_DATA_07_sharpe_and_lintner.xlsx

Lesson 8: ETFs and futures for end users

Futures and ETFs look simple, but an end user faces hidden costs: rolling futures in contango, daily-reset leverage in leveraged ETFs, margin calls and basis risk.

Start with an everyday idea

An ETF is a basket you can buy as a share. A futures contract is an agreement with future settlement. Both can give market exposure, but their cash needs and risks differ.

Words you need

ETF means exchange-traded fund. Tracking error is deviation from a benchmark. Futures leverage means a small collateral balance supports a larger exposure.

Helpful earlier topics: Futures, margin and daily settlement, Portfolio theory and the CAPM

Build the explanation

ETF: a fund that trades like a stock and tracks an index. Low cost, but commodity ETFs usually hold futures, not the commodity.

Roll: futures expire, so a fund sells the near contract and buys the next one every month.

Contango (futures above spot, curve upward): each roll buys higher and the fund slowly loses (negative roll yield). Backwardation (curve downward) gives positive roll yield.

Leveraged ETFs (2×, 3×) reset every day. In a choppy market they lose value even if the index ends flat (volatility decay).

End-user risks: margin/liquidity calls, basis risk, volume (volumetric) risk when you do not know how much you will need to hedge.

Formula and units

Leveraged ETF over 2 days: $(1 + L \cdot r1)(1 + L \cdot r2) \neq 1 + L \cdot (\text{total index return})$

Roll yield (approx): $(F_{near} - F_{next}) / F_{near}$ per roll

L = leverage (2 for a 2× ETF), r = daily index return.

Walk through the sketch

1. An index that goes +10% then -10%: it ends at 99.

2. A 2× daily ETF: +20% then -20%. It ends at 96, worse than "2 × -1%".

3. Contango: futures prices rise with maturity. Each month the fund sells cheap near futures and buys pricier ones.

4. That gap is lost every roll: a slow leak called negative roll yield.

Worked example

An index goes +10% then -10%. What happens to the index and to a 2× daily ETF?

1. Index: 1.10 × 0.90 = 0.99 → down 1%

2. 2× ETF: day 1 = +20%, day 2 = -20%

3. 1.20 × 0.80 = 0.96 → down 4%

4. The ETF lost 4%, not 2× the 1% (2%). That extra loss is volatility decay.

Index -1%, 2× ETF -4%

Common mistakes

1. Expecting a 2× ETF to give 2× the return over months.

2. Ignoring roll cost when comparing a commodity ETF to the spot price.

3. Forgetting that futures hedges can create big cash margin calls even when the hedge works.

Try before checking

Which statement about futures margin is correct?

1. It is the maximum possible loss

2. More cash may be needed after adverse moves

3. It removes all investment risk

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_08_Lecture_7_ETFs_and_Futures_MARKUP.pdf; Futures lecture recording (see "Watch online")

Lesson 9: CDO waterfalls and tranches

A CDO pools many bonds and pays the cash out in order: senior tranche first, then junior, then whatever is left goes to equity. Who gets hurt depends on how many bonds default.

Start with an everyday idea

Think of several cups stacked under a tap. Payments fill the senior cup first. If there is not enough water, the lowest-priority cup goes dry first.

Words you need

A tranche is a slice with a payment priority. Senior is paid first. Equity takes the first losses. A waterfall is the rule for allocating cash and losses.

Helpful earlier topics: Credit risk: default, loss and spreads

Build the explanation

Collateral: the pool of bonds (Mini Project 3: ten $10m bonds, 6% coupons).

Waterfall: each period, pay tranche A what it is owed, then B, then the residual. No carry-forwards in the project version.

Senior (A) is safest; it only loses if many bonds default. Equity/residual takes the first losses.

Correlation matters: with high correlation, bonds tend to default together. That makes "almost nothing" and "lots" of defaults both likelier. Senior tranches get riskier; equity can get safer.

Value each tranche by simulating cash flows (Class 11) and pricing them with Lintner (Class 13).

Formula and units

Pay A = min(cash, owed to A)
Pay B = min(cash - Pay A, owed to B)
Residual = cash - Pay A - Pay B

Repeat every period in every simulation run, then average and discount.

Walk through the sketch

1. Three buckets stacked like a fountain: A on top, B in the middle, residual at the bottom. Cash pours in from the pool.

2. Good period: the pool pays $6m. A fills ($3m), B fills ($2m), $1m spills into the residual.

3. Some defaults: only $4m. A fills, B gets $1m of its $2m, residual gets nothing.

4. Many defaults: $2m. Even A comes up short. B and residual get nothing.

5. Rule: losses hit from the bottom up. Senior is protected by everyone below it.

Worked example

This period A is owed $3m and B is owed $2m. The pool pays (a) $6m, (b) $4m, (c) $2m. Who gets what?

1. (a) $6m: A gets 3, B gets 2, residual gets 1

2. (b) $4m: A gets 3, B gets 1 (short by 1), residual 0

3. (c) $2m: A gets 2 (short by 1), B 0, residual 0

4. So the residual is hit first, then B, and A only in bad cases.

(a) 3/2/1 (b) 3/1/0 (c) 2/0/0

Common mistakes

1. Paying tranches pro rata instead of in order.

2. Forgetting LGD: a defaulted bond still pays its recovery.

3. Assuming correlation always makes every tranche riskier.

Try before checking

Which tranche normally absorbs the first losses?

1. Senior

2. Equity

3. All equally

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_REVIEW_04_Mini_Project_3_CDO_Risk_Part_1_Instructions.txt; FRE6103_NOTES_01_Project_and_recording_notes.md; FRE6103_LECTURE_10_Lecture_03_Corporate_Bonds_and_CDOs.pdf

Lesson 10: Volatility smiles and implied densities

If you back out σ from real option prices, it is not the same for every strike. That pattern tells you what the market really thinks future prices look like.

Start with an everyday idea

Ask the option-pricing recipe a backward question: which volatility would make its price match the market price? Different strikes can give different answers.

Words you need

Implied volatility is inferred from an option price. A smile is the pattern across strikes. A surface adds time to expiry. An implied density describes pricing probabilities, not a guaranteed forecast.

Helpful earlier topics: Black-Scholes and the Greeks

Build the explanation

Implied volatility: the σ you plug into Black-Scholes to match a market price. Solve it with Goal Seek.

Smile / skew: plot implied vol against strike. For stocks, low strikes usually have higher vol (a skew): crashes are feared more than the lognormal model allows.

Volatility surface: implied vol across strike and maturity.

Implied density (Breeden-Litzenberger): the second derivative of call price with respect to strike gives the market's risk-neutral probability density. In practice: a butterfly spread.

European vs Asian: an Asian option pays on the average price, which is less volatile, so it is cheaper. It usually needs simulation.

Formula and units

Implied density: $q(K) = e^{(rT)} \times \partial^2 C/\partial K^2$

Butterfly approx: $q(K) \approx e^{(rT)} \times [C(K-h) - 2 \cdot C(K) + C(K+h)] / h^2$

C(K) = call price at strike K, h = gap between strikes.

Walk through the sketch

1. Black-Scholes assumes one volatility for all strikes: a flat line.

2. Real markets show a skew: low strikes (crash protection) have much higher implied vol.

3. A butterfly (buy 95, sell two 100s, buy 105) pays a little tent. Its cost measures probability near 100.

4. Line up many butterflies and you get the implied density. Compared with the lognormal (dashed), the left tail is fatter.

Worked example

Calls cost 8.00, 5.00 and 2.80 at strikes 95, 100, 105. Estimate the probability density at 100 (ignore discounting). Roughly what is the chance S_T ends between 97.5 and 102.5?

1. Butterfly cost: 8.00 - 2 × 5.00 + 2.80 = 0.80

2. Divide by $h^2 = 25$: density ≈ 0.032 per $1

3. Over a $5 window: 0.032 × 5 = 0.16

4. So about a 16% risk-neutral chance of ending near $100.

Density ≈ 0.032 per $; about 16% chance

Common mistakes

1. Reading a risk-neutral density as a real-world forecast.

2. A negative butterfly cost means arbitrage or bad data, not a negative probability.

3. Using one σ for every strike when the smile is steep.

Try before checking

Implied volatility comes from…

1. Only past stock returns

2. An observed option price and a pricing model

3. A guaranteed future price

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_15_Volatility_Surface.pdf; FRE6103_LECTURE_09_Lecture_9_BSM_and_Implied_Densities.pdf; FRE6103_DATA_01_European_and_Asian_option_benchmarking.xlsx

Lesson 11: Real options and the Merton model

Choices in a business (wait, expand, abandon) are options, so they have value that plain NPV misses. Even a company's stock is an option on its assets.

Start with an everyday idea

A business can wait before building a factory, just as an option holder can wait before exercising. Flexibility can have value when the future is uncertain.

Words you need

A real option is flexibility over a business decision. Merton models a company’s equity like a call on its assets, with debt due at a future date.

Helpful earlier topics: The four derivatives and their payoffs, Credit risk: default, loss and spreads, Black-Scholes and the Greeks

Build the explanation

Real option: the right, not the duty, to take a business action later. Waiting lets you invest only if things go well.

Value it like Class 16: build a tree of the project value, find risk-neutral p, roll back.

If the option value is bigger than investing now (NPV today), wait.

Energy example (VL14): a power plant is a call on the spread between power and fuel prices: run it only when profitable.

Merton model: equity = call option on the firm's assets with strike = debt. Default happens if assets end below the debt. Distance to default counts how many σ the firm is from that line.

Formula and units

Risk-neutral p = [V0·(1 + rf) - V_down] / (V_up - V_down)

Option value = [p·max(V_up - I, 0) + (1-p)·max(V_down - I, 0)] / (1 + rf)

Merton: Equity = V·N(d1) - D·e^(-rT)·N(d2)

DD = [ln(V/D) + (μ - σ²/2)T] / (σ√T), PD = N(-DD)

V = value of project or firm assets, I = investment cost, D = face value of debt.

Walk through the sketch

1. Today the project is worth $105 and costs $100. NPV = $5.

2. Next year it is worth $140 (good) or $80 (bad).

3. If you wait, you invest only in the good case: gain $40. In the bad case you walk away: $0.

4. Rolled back with p = 0.504: the right to wait is worth $19.21. Waiting wins.

5. Same idea for a firm: shareholders get max(Assets - Debt, 0). Equity is a call on the assets.

Worked example

A project is worth $105 today and costs $100. Next year it will be worth $140 or $80; rf = 5%. Invest now, or keep the right to invest next year (cost still $100)?

1. NPV now = 105 - 100 = 5

2. p = (105 × 1.05 - 80) / (140 - 80) = 30.25/60 = 0.504

3. Wait: next year invest only if up: payoff 40 or 0

4. Option value = 0.504 × 40 / 1.05 = 19.21. Bigger than 5, so wait.

5. Merton check: V = 100, D = 80, μ = 6%, σ = 25%, T = 1 → DD = (0.2231 + 0.0288)/0.25 = 1.01, PD = N(-1.01) ≈ 15.7%

Waiting is worth $19.21 vs $5 now. Merton PD ≈ 15.7%

Common mistakes

1. Using the real-world probability in the roll-back.

2. Comparing the option with zero instead of with investing now.

3. In Merton, using the stock's volatility as the asset volatility.

Try before checking

A factory project can wait one year for more information. That flexibility…

1. May have value

2. Must be worthless

3. Guarantees profit

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf; FRE6103_LECTURE_14_VL14_Real_Options_in_Energy.pdf; FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf; Risk syllabus RMN6 (Merton, distance to default)

Paper lab: Move money through time

A promised $121 arrives in two years. The annual rate is 10%. Move it backward one year at a time.



Predict and try

1. Predict: should today’s value be greater or smaller than $121?

2. Compute PV for rates 0%, 5% and 10%. Keep the two-year date fixed.

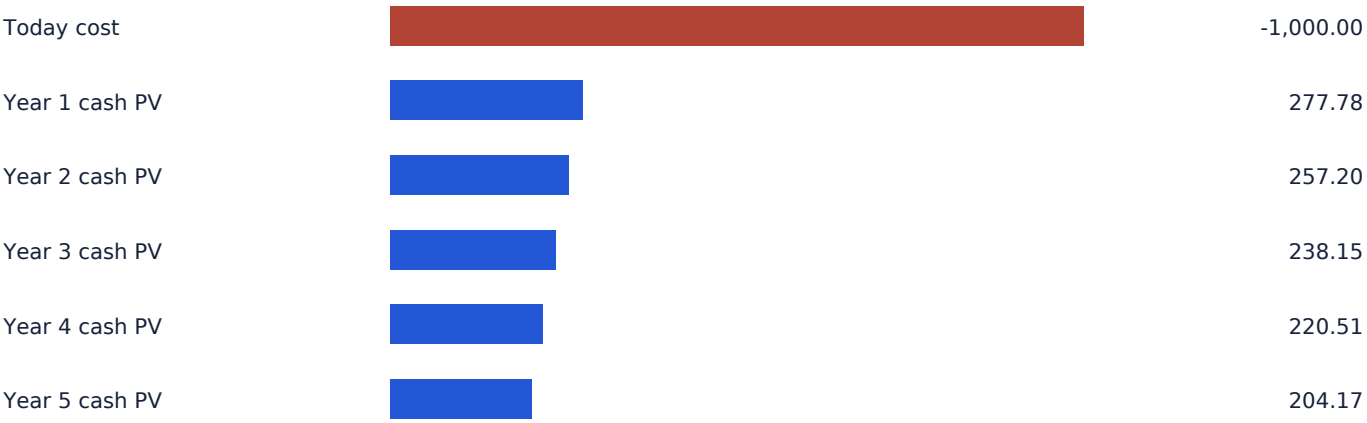
Compare with the explanation

PV = 121/(1+r)^2: $121.00 at 0%, $109.75 at 5%, $100.00 at 10%. A higher discount rate lowers PV. Check: $100 × 1.1 × 1.1 = $121.

Model assumptions: Known future payment; annual compounding; same rate for both years.

Paper lab: Find the rate at which a project breaks even

Pay $1,000 now and receive $300 at each year-end for five years. Discount at 8%.



Predict and try

1. Compute the PV of each future payment and subtract the initial cost.

2. Raise the rate to 20%. What changes? What happens for a one-year cost of $100 and payment of $100 or $50?

Compare with the explanation

At 8%, PV of inflows = $1,197.81 and NPV = +$197.81. At 20%, NPV = -$102.82. The five-year project’s IRR is approximately 15.2382%. The one-year examples have IRRs of 0% and -50%, respectively. Zero NPV means exactly earning the required rate; a negative IRR is a valid losing return.

Model assumptions: Equal end-year payments, one initial outflow, positive later inflows. This cash-flow pattern has a unique IRR greater than -100%; more complex patterns can behave differently.

Paper lab: Change the yield while coupons stay fixed

A bond pays a $50 coupon annually for five years and repays $1,000 with its final coupon.



Predict and try

1. Predict the price relative to face value when yield equals the 5% coupon rate.

2. At 7%, discount all five coupons and the $1,000 principal. Explain why the price falls.

Compare with the explanation

Prices are $1,091.59 at 3%, $1,000.00 at 5%, and $918.00 at 7%. Higher yield discounts the unchanged cash flows more heavily. The last payment is $1,050, not just the $50 coupon.

Model assumptions: Annual coupons, flat yield, no default, valuation immediately after a coupon date.

Paper lab: Build the required return from two pieces

The risk-free rate is 4%; the expected market return is 9%. The market risk premium is 5 percentage points.



Predict and try

1. Calculate required returns for beta 0, 1 and 1.2.

2. Does beta 0 mean the stock has no risk of its own?

Compare with the explanation

E[R] = 4% + beta × (9% - 4%). Returns are 4%, 9% and 10% for beta 0, 1 and 1.2. Beta 0 means zero covariance with the market in this model, not zero total risk. Diversifiable company-specific risk can still exist.

Model assumptions: One-period CAPM with specified inputs. Expected returns are not guaranteed realized returns.

Lesson-check answers

1. Money today vs money later

Answer: $100. Divide $110 by 1.10 to undo one year of growth: $100. Multiplying would move money forward in time.

2. Loans: payments, default and prepayment

Answer: $85. Start with $100, add $5 interest, subtract the $20 payment: $85 remains.

3. Bonds and building a zero curve

Answer: $100. There is one payment: 110 ÷ 1.10 = 100. More payment dates mean more discounted amounts to add.

4. Capital budgeting: NPV and IRR

Answer: $0. PV = 110 ÷ 1.10 = 100. Subtract the $100 cost: NPV = 0.

5. Simulation, correlation and Cholesky

Answer: To represent how assets move together. Independent draws would miss co-movement. Correlation matters for the risk of the combined portfolio.

6. Random walks and stock price paths

Answer: They use different random shocks. The same model can produce many paths because each path receives different random shocks.

7. Portfolio theory and the CAPM

Answer: 13%. Required return = 3% + 2 × 5% = 13%. Beta multiplies the premium, not the risk-free rate.

8. ETFs and futures for end users

Answer: More cash may be needed after adverse moves. Margin is collateral. Adverse daily settlement can require extra cash; losses are not capped at the initial margin.

9. CDO waterfalls and tranches

Answer: Equity. Equity is the first-loss slice. Senior protection can still be exhausted by sufficiently large losses.

10. Volatility smiles and implied densities

Answer: An observed option price and a pricing model. You solve backward from price to volatility. Historical volatility instead measures past variation.

11. Real options and the Merton model

Answer: May have value. Waiting can avoid a bad investment, though delay can also lose revenue. The value depends on assumptions.

Valuation midterm-style practice

7 questions | 100 points | Learn at your own pace.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Derive a cash-flow value (15 points)
2. Build a Lintner CAPM table (15 points)
3. Correlate and uncorrelate random draws (15 points)
4. Tell uncertainty from simulation noise (10 points)
5. Critique a correlation claim (15 points)
6. Write a process with reversion and jumps (15 points)
7. Explain capital and performance (15 points)

Question 1: Derive a cash-flow value

15 points | 2025A Q1 · MBE/GVE and cash-flow series

Payments starting one period from now are 0, 2, 6, 12, 20, 30, … forever. Each payment increases by 2 more than the previous increase. Derive the present value V at a constant per-period discount rate r > 0. Show the shift-and-subtract argument, not just a memorized answer. Then value the stream at r = 10%.

Show your setup, units, calculation and interpretation.

Question 2: Build a Lintner CAPM table

15 points | 2025A Q2 · Cash-flow covariance and valuation

Two securities have expected next-period cash flows 40 and 60. Their covariance matrix is [[64,16],[16,144]]. The risk-free rate is zero. Asset 1 is worth 36. Using Vi = E[Xi] - A Cov(Xi, XM) with XM = X1 + X2, find A, asset 2’s value, total market value, value weights, expected returns and return standard deviations.

Show your setup, units, calculation and interpretation.

Question 3: Correlate and uncorrelate random draws

15 points | 2025A Q3 · Cholesky and matrix orientation

Find a lower-triangular Cholesky factor of Σ = [[25,15],[15,45]]. If each row of an n×2 matrix U contains independent standard-normal draws, how do you obtain correlated zero-mean rows W? How do you undo the transformation?

Show your setup, units, calculation and interpretation.

Question 4: Tell uncertainty from simulation noise

10 points | 2025A Q4 · Standard error and sample-size scaling

A simulation has n = 400 and maxerr = 4.00. Use the supplied exam key’s convention maxerr = 2 × SE. Find SE and sample standard deviation s. How many independent draws are needed for maxerr at most 1.50, assuming s stays the same?

Show your setup, units, calculation and interpretation.

Question 5: Critique a correlation claim

15 points | 2025A Q5 · Reasoning from data

An analyst shows two asset-price series rising through a year and reports a level correlation of +0.65. They conclude the assets respond to the same economic forces and provide little diversification. Write a short critique and describe a better analysis. You are not given the return data.

Show your setup, units, calculation and interpretation.

Question 6: Write a process with reversion and jumps

15 points | 2025A Q6 · Translate a story into a model

Write a continuous-time price model with mean reversion toward μ at speed κ, diffusion volatility σ and upward jumps arriving at intensity π. At each jump, size J is exponential with rate λ with probability 40%, or a fixed $3 with probability 60%. Give an Euler simulation step and explain all random draws.

Show your setup, units, calculation and interpretation.

Question 7: Explain capital and performance

15 points | 2025A Q7 · Short written explanation

Explain why capital differs from cash, why a bank needs a loss-absorbing cushion, and why comparing profit alone can mislead when activities take different risks. State how the supplied course key describes risk-adjusted performance, and distinguish that from common return-on-economic-capital terminology.

Show your setup, units, calculation and interpretation.

Solution 1: Derive a cash-flow value

First, the words

A perpetuity is a payment stream with no last payment. GVE relates today's value to the next payment and the value remaining afterward. Here r is a positive rate per period.

A smaller problem first

For $2 every period forever, starting next period, value is 2/r. At 10%, that is $20. Why? The first payment plus the remaining value, discounted once, must equal today's value: V = (2 + V)/(1+r).

Hint

Write V for the original stream and Vnext for the remaining 2, 6, 12, 20, … stream. The difference Vnext - V is 2, 4, 6, … . First value that simpler increasing stream.

Work through it

1. Let L be the value of 2,4,6,… . After one period its remaining stream is 4,6,8,…, worth Lnext. The difference Lnext - L is a flat $2 perpetuity, worth 2/r.

2. Apply GVE to L: $(1+r)L = 2 + Lnext = 2 + L + 2/r$. Therefore $L = 2(1+r)/r^2$.

3. Now let V be the value of 0,2,6,12,… . Its first payment is zero, so $(1+r)V = Vnext$. The shifted stream minus the original stream is 2,4,6,…, so Vnext - V = L.

4. Subtract V from the GVE: $rV = L$. Hence $V = 2(1+r)/r^3$.

5. At r = 0.10, $V = 2 \times 1.10 / 0.001 = \$2{,}200$. Timing matters: the first payment at t = 1 is zero. The infinite series converges for r > 0.

Mark your reasoning

1. 4: timeline and GVE for the original stream

2. 5: values the first-difference stream

3. 4: formula and $2,200

4. 2: r > 0 and first-payment timing

Shimko, Valuation Principles Chapters 1-2: MBE/GVE and growing payment patterns. This follows the supplied triangular-series question with scaled payments.

My mark: ____ / 15. My next correction: _____

Solution 2: Build a Lintner CAPM table

First, the words

Mean is the expected future cash flow. Variance is a measure of spread; covariance measures how two cash flows vary together. A is the course model's price per unit of covariance risk.

A smaller problem first

With no risk-free interest, value = expected cash flow - risk charge. If the mean is $50 and the charge is $6, value is $44. This question uses the course's cash-flow CAPM equation, not just a beta/return shortcut.

Hint

Cov(X1,XM) is the sum of row 1, not just asset 1's variance. Return standard deviation is cash-flow standard deviation divided by today's value.

Work through it

1. Row sums: Cov(X1,XM) = 64 + 16 = 80; Cov(X2,XM) = 16 + 144 = 160.

2. Use asset 1: 36 = 40 - 80A, so A = 4/80 = 0.05. Asset 2 value = 60 - 0.05 × 160 = 52.

3. Market value M = 36 + 52 = 88. Value weights are 36/88 = 40.91% and 52/88 = 59.09%.

4. Expected returns: 40/36 - 1 = 11.11%; 60/52 - 1 = 15.38%; market = 100/88 - 1 = 13.64%.

5. Cash-flow standard deviations: √64 = 8 and √144 = 12. Divide by values: 8/36 = 22.22%, 12/52 = 23.08%.

6. Market cash-flow variance is the sum of all four matrix entries: 240. Its return standard deviation is √240 / 88 = 17.60%.

Mark your reasoning

1. 4: row sums and A = 0.05

2. 3: values and weights

3. 4: expected returns

4. 4: standard deviations including market covariance

Valuation Lecture 6: MPT and the CAPM; the supplied Sharpe/Lintner spreadsheet and the 2025A key. This is a course-specific one-period setup.

My mark: ____ / 15. My next correction: _____

Solution 3: Correlate and uncorrelate random draws

First, the words

A matrix is a rectangular table of numbers. Transpose (T) swaps rows and columns. An inverse reverses a transformation. Cholesky finds a lower-triangular matrix C with $\Sigma = CC^T$.

A smaller problem first

For a lower-triangular 2×2 matrix C = [[a,0],[b,c]], multiplying C by C^T gives $[[a^2,ab],[ab,b^2+c^2]]$. Match these entries to the covariance matrix one at a time.

Hint

Start with $a^2 = 25$. Then $ab = 15$. Finish with $b^2 + c^2 = 45$. Pay attention to whether observations are stored in rows or columns.

Work through it

1. $a = \sqrt{25} = 5$. $b = 15/5 = 3$. $c = \sqrt{(45 - 3^2)} = \sqrt{36} = 6$.

2. $C = [[5,0],[3,6]]$. Check $CC^T = [[25,15],[15,9+36]] = \Sigma$.

3. For observations in rows, $W = UC^T$. Row transformation gives the required covariance.

4. Undo it with $U = W(C^T)^{-1}$. If W has nonzero means, subtract those means first.

5. A valid covariance matrix must be positive semidefinite. This example is positive definite; the diagonal square roots are positive and the inverse exists.

Mark your reasoning

1. 7: factor entries

2. 3: multiplication check

3. 3: forward transformation

4. 2: inverse and orientation

Shimko simulation excerpt: multivariate simulation and Cholesky; Valuation Lecture 4 and Cholesky spreadsheet.

My mark: ____ / 15. My next correction: _____

Solution 4: Tell uncertainty from simulation noise

First, the words

Standard deviation (s) describes spread in individual simulated outcomes. Standard error (SE) describes uncertainty in their average. More draws reduce SE, not the underlying risk of each outcome.

A smaller problem first

SE = s/√n. If s = 20 and n = 100, then SE = 2. Four times as many independent draws cut the standard error in half.

Hint

First halve maxerr. Then multiply SE by √n. Sample size grows with the square of the ratio of old to new error.

Work through it

1. SE = 4/2 = 2.00.

2. s = SE × √400 = 2 × 20 = 40.00.

3. nnew = 400 × (4/1.5)² = 2,844.44…. Round upward to 2,845 draws.

4. The factor 2 is the convention used here, approximately a 95% normal interval. Dependence or poor simulation assumptions can invalidate this simple error estimate.

Mark your reasoning

1. 3: SE = 2

2. 3: s = 40

3. 3: 2,845 draws, rounded up

4. 1: independent-draw and unchanged-spread assumptions

Valuation Lecture 4: Simulation; supplied 2025A Q4/key. Its numbers show the maxerr = 2 SE convention.

My mark: ____ / 10. My next correction: _____

Solution 5: Critique a correlation claim

First, the words

A price level is a price at a date. A return measures the change relative to the previous price. A shared trend can create a strong level correlation without synchronized short-term changes.

A smaller problem first

Two people can both walk uphill while one pauses whenever the other speeds up. Their heights can rise together even when their step-by-step changes do not.

Hint

Separate what the chart shows, what it does not establish, and what you would calculate next. Do not invent a return correlation.

Work through it

1. The correlation is between levels, which can be persistent or trending. Shared trends can drive the statistic without establishing a causal relationship.

2. Calculate aligned period-by-period returns or changes, choose an economically relevant frequency, and inspect their scatter plot and correlation.

3. Check missing dates, outliers, sample length, changing relationships and serial dependence. Returns are not automatically independent or identically distributed.

4. Diversification depends on the joint behavior of portfolio returns, exposures and the horizon. A single level-correlation number is insufficient.

5. We cannot infer the sign or magnitude of return correlation without the actual data. State that limitation rather than claiming it must be negative.

Mark your reasoning

1. 5: identifies level/trend problem

2. 4: proposes returns/changes analysis

3. 3: limits causal/diversification claim

4. 3: acknowledges missing data and model checks

Supplied 2025A Q5/key; Valuation stochastic-process and simulation materials. The particular sign discussed in the source key is not a universal rule.

My mark: ____ / 15. My next correction: _____

Solution 6: Write a process with reversion and jumps

First, the words

Mean reversion pulls the price toward a long-run level μ. κ controls the speed. Brownian shocks model small random moves. A Poisson process N counts occasional jump arrivals with intensity π per unit time.

A smaller problem first

If the current price is 80 and the long-run level is 100, κ(100 - 80) is positive: the drift points upward. If the price is 120, that drift points downward.

Hint

Separate whether a jump occurs from how large that jump is. Brownian increments scale with √Δt; drift scales with Δt.

Work through it

1. A compound-Poisson version is dS = κ(μ - S)dt + σdW + J dN, with independently sampled marks J at each jump and N a Poisson counting process of intensity π.

2. For each jump, draw B from Bernoulli(0.4). If B = 1, use J = -ln(U)/λ with U uniform on (0,1); otherwise use J = 3. λ must be positive.

3. For one simulation step, draw Z standard normal and K Poisson(πΔt), independently. Draw K jump sizes J1,...,JK from the mixture.

4. Euler step: Snext = S + κ(μ - S)Δt + σ√Δt Z + Σj Jj. If K = 0 the sum is zero.

5. For very small πΔt, a Bernoulli jump indicator with probability approximately πΔt is a small-step approximation. It misses multiple arrivals; do not use it as an exact large-step rule.

6. This model has upward jumps and an additive diffusion, which can permit negative prices. Mean reversion is the drift component; jumps shift the overall stationary mean unless compensated. State assumptions.

Mark your reasoning

1. 4: mean-reverting drift and diffusion

2. 3: arrival process

3. 4: mixture and exponential sampling

4. 4: time scaling and separate independent draws

Valuation Lecture 5: Stochastic Processes; Hull 8e Chapter 13 for Wiener-process foundations. The source key has compressed/inconsistent symbols; this practice uses explicit μ for mean and π for jump intensity.

My mark: ____ / 15. My next correction: _____

Solution 7: Explain capital and performance

First, the words

Cash is a liquid asset used to pay bills. Capital is a loss-absorbing funding cushion; economic capital is a model-based amount set aside against risk. They are not interchangeable.

A smaller problem first

A bank can hold many valuable long-term assets but little cash today. It can also have plenty of cash financed by debt and very little equity. Liquidity and loss absorption are different questions.

Hint

Write one sentence each for definition, banking purpose, comparable risk-adjusted performance and metric conventions.

Work through it

1. Cash concerns immediate payment ability. Capital absorbs losses and supports risk-taking; economic capital estimates a cushion for adverse losses under a chosen model and confidence level. It is not a guaranteed maximum-loss bound.

2. Banks need loss-absorbing resources because credit, market and operational losses can otherwise threaten creditors and continued operation.

3. Two desks with the same profit may use very different amounts of risk capital. Compare performance after accounting for funding and risk, using consistent definitions and time horizons.

4. The supplied 2025A key names the Sharpe ratio or dollarized Sharpe ratio. Learn that course wording: the usual Sharpe ratio measures excess return per unit of return standard deviation.

5. Common industry usage also includes RAROC/return on economic capital, which uses a capital denominator with a specified income definition. These measures are related ideas, not identical formulas. State the convention requested by the instructor.

Mark your reasoning

1. 4: cash versus capital

2. 4: loss absorption and bank risk

3. 4: risk-adjusted comparison

4. 3: accurate course wording and distinction between metrics

Supplied 2025A Q7/key; Risk syllabus and Valuation capital/performance discussion. Ask the instructor to confirm metric terminology for the current assessment.

My mark: ____ / 15. My next correction: _____

Valuation mock final

8 questions | 100 points | 120 minutes suggested practice time.

Original practice, not an official final or a prediction of exam scope. Confirm current course rules. Questions come first; worked solutions and marking guides follow all questions. The suggested time and point weights for mock finals are study choices.

1. Derive an increasing cash-flow value (12 points)
2. Bootstrap and check interest-rate sensitivity (12 points)
3. Price an option using a cash-flow risk adjustment (12 points)
4. Audit a multivariate simulation (12 points)
5. Read a probability clue from option prices (12 points)
6. Compare committing now with waiting (14 points)
7. Connect financing assumptions to capital budgeting (14 points)
8. Trace losses and choose a risk response (12 points)

Question 1: Derive an increasing cash-flow value

12 points | Revised syllabus week 1; historical midterm derivation style.

Cash flows at years 1, 2, 3, ... are $10, $20, $30, ... forever. Derive V using a shift and GVE, then evaluate it at r = 10% annually. Explain why using the same rate for uncertain future cash flows needs an additional assumption.

Show your setup, units, calculation and interpretation.

Question 2: Bootstrap and check interest-rate sensitivity

12 points | Revised syllabus weeks 2-3: bonds, zero curve and interest-rate risk.

The one-year annual spot rate is 4%. A two-year annual-coupon bond with face $100 and coupon 5% costs $100. Derive D2 and the two-year annually compounded spot rate. Separately a bond portfolio is worth $20,000 with modified duration 4.2. Estimate its value change if its yield rises by 0.50 percentage points. State what you are ignoring.

Show your setup, units, calculation and interpretation.

Question 3: Price an option using a cash-flow risk adjustment

12 points | Revised syllabus weeks 6 and 9: Lintner CAPM and option valuation.

There are two equally likely physical states. Market cash flow XM is 80 or 120; asset cash flow X is 70 or 130 in the same states. A call pays Y = max(X - 100, 0). Use V(Y) = [E(Y) - A Cov(Y,XM)]/(1 + rf), with A = 0.01 and rf = 5%. Find the call’s expected cash flow, covariance, risk-adjusted value and value if the covariance charge were wrongly omitted. Explain the difference.

Show your setup, units, calculation and interpretation.

Question 4: Audit a multivariate simulation

12 points | Revised syllabus weeks 4-5; historical midterm computational style.

Find a lower-triangular C for covariance matrix [[9,6],[6,20]]. State the transformation from independent standard-normal rows U to correlated rows W and its inverse. A separate simulation gives payoff standard deviation s = 24 from n = 144 independent draws. With maxerr = 2SE, find its maxerr and the number of draws needed for maxerr no greater than 1, keeping s unchanged.

Show your setup, units, calculation and interpretation.

Question 5: Read a probability clue from option prices

12 points | Revised syllabus week 10: implied probability distributions.

Same-expiry European calls on a non-dividend stock have prices C(90) = 16, C(100) = 9, C(110) = 4. The expiry discount factor is D = 0.95. Find the cost of the 90/100/110 butterfly, its maximum terminal payoff, and a centered finite-difference estimate of the risk-neutral density near strike 100 using $fQ(K) \approx [C(K-h) - 2C(K) + C(K+h)]/(D\,h^2)$, h = 10. Interpret fQ × h carefully. What fails if C(100) were 11 instead?

Show your setup, units, calculation and interpretation.

Question 6: Compare committing now with waiting

14 points | Revised syllabus week 11: contracts and real options.

Next year a project asset will be worth 160 or 80. Treat these as market-consistent asset values with risk-neutral probabilities 0.5 each. The annual risk-free growth factor is 1.05. You can either pay 100 now to own the asset next year, or hold a free right to pay 100 next year after seeing its value. Find immediate-investment NPV, the value of waiting and the incremental flexibility value. Explain why using this method for a nontraded business needs care.

Show your setup, units, calculation and interpretation.

Question 7: Connect financing assumptions to capital budgeting

14 points | Revised syllabus weeks 11-12: capital budgeting and capital structure.

A company has market equity value 600 and debt 400. Its cost of equity is 12%, pre-tax cost of debt is 6%, and corporate tax rate is 25%. Under a stable target financing mix and comparable project risk, calculate after-tax WACC. A project costs 100 now and pays one unlevered after-tax cash flow of 109 next year. Find NPV and IRR, and explain the decision at this rate. Why can you not keep the same component costs while increasing debt indefinitely to make value rise?

Show your setup, units, calculation and interpretation.

Question 8: Trace losses and choose a risk response

12 points | Revised syllabus: corporate bonds/CDOs and later corporate risk management.

A $100 debt pool has equity 5, mezzanine 10 and senior 85. Allocate a realized $12 pool loss. Separately a company’s annual loss distribution is 0 with probability 90%, 10 with probability 8%, and 40 with probability 2% (all in $m). Find expected loss, 95% VaR as the smallest loss with cumulative probability at least 95%, and ES as the average of the worst 5%, splitting the mass at the VaR threshold as needed. Give one reason management should examine stress scenarios as well.

Show your setup, units, calculation and interpretation.

Solution 1: Derive an increasing cash-flow value

First, the words

GVE links today's value to the next cash flow and remaining value. A perpetuity continues indefinitely; the discount rate here is positive and per year.

A smaller problem first

A flat $10 perpetuity beginning next year is worth 10/r. Use a shifted stream to handle payments that grow by a fixed dollar amount.

Hint

After the first year, the remaining stream is the original stream plus $10 each year.

Work through it

1. Let Vnext value the stream 20,30,40,... one year later. Relative to 10,20,30,... it adds a $10 perpetuity, so Vnext = V + 10/r.

2. GVE: (1 + r)V = 10 + Vnext = 10 + V + 10/r.

3. Therefore rV = 10 + 10/r and V = 10(1 + r)/r^2.

4. At r = 0.10, V = 10 × 1.1/0.01 = $1,100. The first payment is at year 1 and r > 0 ensures convergence.

5. For uncertain cash flows, their risk and timing must be priced consistently. A risk-free rate applied to raw expected risky cash flows generally omits a risk adjustment.

Mark your reasoning

1. 4: shifted stream

2. 3: GVE and formula

3. 3: value, timing and domain

4. 2: risk limitation

Shimko Valuation Principles Chapters 1-2 and Lecture 1.

My mark: ____ / 12. My next correction: _____

Solution 2: Bootstrap and check interest-rate sensitivity

First, the words

Bootstrapping finds each discount factor from traded cash flows. Duration is a first-order approximation, not an exact prediction for large yield moves.

A smaller problem first

A one-year zero paying $100 at 4% annually costs 100/1.04. That gives D1 = 1/1.04.

Hint

Price = 5D1 + 105D2. For duration, convert 0.50 percentage points to 0.005.

Work through it

1. D1 = 1/1.04 = 0.96153846.

2. 100 = 5D1 + 105D2, so D2 = (100 - 5/1.04)/105 = 0.90659341.

3. Two-year spot rate z2 = D2^(-1/2) - 1 = 5.02525% annually.

4. Approximate change = -modified duration × value × yield change = -4.2 × 20,000 × 0.005 = -$420; estimated new value $19,580.

5. The approximation ignores convexity and changes in cash flows; a single yield shift also does not capture arbitrary curve twists or spread changes.

Mark your reasoning

1. 4: bootstrapping setup

2. 3: D2 and spot rate

3. 3: duration calculation

4. 2: limitations

Valuation bond/interest-rate lecture; zero-coupon term-structure mini-project topics.

My mark: ____ / 12. My next correction: _____

Solution 3: Price an option using a cash-flow risk adjustment

First, the words

An option’s cash flow is nonlinear. Compute its own covariance with the market before applying the supplied cash-flow pricing model.

A smaller problem first

A call paying max(X - 100, 0) on asset cash flows 70 and 130 pays 0 and 30, not -30 and 30.

Hint

Covariance is the average of the products of deviations from each variable’s mean.

Work through it

1. Y is 0 in the low state and 30 in the high state. E(Y) = 15; E(XM) = 100.

2. Cov(Y,XM) = 0.5(-15)(-20) + 0.5(15)(20) = 300.

3. Risk charge = 0.01 × 300 = 3. Call value = (15 - 3)/1.05 = $11.4286.

4. Ignoring risk gives 15/1.05 = $14.2857, an overstatement of $2.8571 in this stipulated model.

5. The option pays in the high-market state, so the model charges for positive market covariance. This is a given one-period model, not an assertion that physical probabilities equal risk-neutral probabilities.

Mark your reasoning

1. 2: payoffs and mean

2. 4: covariance

3. 3: adjusted price

4. 3: comparison and interpretation

Valuation Lecture 6 and revised syllabus week 9; follow your instructor’s notation and calibration.

My mark: ____ / 12. My next correction: _____

Solution 4: Audit a multivariate simulation

First, the words

Cholesky creates correlated draws. Standard error measures uncertainty in an estimated mean; it is different from the standard deviation of individual outcomes.

A smaller problem first

For $C = [[a,0],[b,c]]$, $CC^T = [[a^2,ab],[ab,b^2+c^2]]$. With observations in rows, transform independent draws using C^T.

Hint

Match a^2, ab and $b^2 + c^2$. Then use SE = s/sqrt(n).

Work through it

1. a = 3, b = 2, c = 4. Thus $C = [[3,0],[2,4]]$. Check $CC^T = [[9,6],[6,20]]$.

2. For row observations $W = UC^T$; recover $U = W(C^T)^{-1}$, after subtracting means if needed.

3. SE = 24/sqrt(144) = 2. Therefore maxerr = 4.

4. Need $2 \times 24/\sqrt{n} \leq 1$, so $n \geq 48^2 = 2{,}304$.

5. Independence and unchanged outcome spread are assumptions. More draws reduce sampling error, not model bias or the underlying payoff risk.

Mark your reasoning

1. 4: Cholesky and check

2. 3: row transformation and inverse

3. 3: error and sample size

4. 2: limitations

Valuation simulation lecture and Cholesky spreadsheet; historical maxerr convention is specified explicitly.

My mark: ____ / 12. My next correction: _____

Solution 5: Read a probability clue from option prices

First, the words

A butterfly combines calls at three equally spaced strikes. Call-price curvature contains information about risk-neutral probabilities, not directly real-world probabilities.

A smaller problem first

A long butterfly buys the low-strike call, sells two middle-strike calls and buys the high-strike call. Its payoff is never negative.

Hint

Positive curvature is required by no arbitrage. A density has units of probability per dollar.

Work through it

1. Butterfly cost = 16 - 2 × 9 + 4 = $2.

2. The payoff rises from 0 at S = 90 to $10 at S = 100 and falls to 0 at S = 110; outside that range it is zero.

3. Density estimate = 2/(0.95 × 100) = 0.0210526 per dollar.

4. Multiplying by a $10 interval gives roughly 0.2105, or 21.05%, only if a local constant-density approximation is reasonable. This coarse estimate is not an exact interval probability.

5. If the middle call were 11, butterfly cost would be -$2 while terminal payoff remains nonnegative. That violates convexity and allows an arbitrage under frictionless trading.

6. These are pricing-measure probabilities. Risk premia and the finite strike spacing prevent reading them as a precise real-world forecast.

Mark your reasoning

1. 3: cost and payoff

2. 3: density calculation

3. 3: careful interpretation

4. 3: convexity violation

Valuation implied-density and volatility-surface lectures; the formula is supplied to make the exercise self-contained.

My mark: ____ / 12. My next correction: _____

Solution 6: Compare committing now with waiting

First, the words

An option to invest lets you reject a bad outcome. A risk-neutral probability is given here; do not substitute physical forecasts without changing the model.

A smaller problem first

If next year’s project asset is worth 80 and investment costs 100, an optional investment pays 0 because you can decline.

Hint

Discount the asset value for the commit-now case; discount max(asset value - investment cost, 0) for the waiting case.

Work through it

1. Asset PV = (0.5 × 160 + 0.5 × 80)/1.05 = $114.2857.

2. Immediate-investment NPV = 114.2857 - 100 = $14.2857.

3. Waiting payoffs are $60 in the high state and $0 in the low state.

4. Waiting value = (0.5 × 60 + 0.5 × 0)/1.05 = $28.5714.

5. Incremental value relative to committing now = 28.5714 - 14.2857 = $14.2857. It includes the ability to avoid bad investment and defer the expenditure.

6. For a nontraded project, replication may be unavailable. Justify the pricing probabilities/risk adjustment, possible competition and any cost of waiting. The supplied assumptions make this toy model calculable.

Mark your reasoning

1. 3: asset PV and immediate NPV

2. 4: contingent payoffs and option value

3. 3: incremental comparison

4. 4: assumptions and nontraded limitation

Valuation real-options-in-energy lecture; Welch capital budgeting and option intuition.

My mark: ____ / 14. My next correction: _____

Solution 7: Connect financing assumptions to capital budgeting

First, the words

WACC weights financing costs by market values. The tax adjustment here assumes deductible interest and a usable corporate tax shield.

A smaller problem first

If 60% of funding is equity and 40% is debt, multiply each financing cost by its weight; do not average them equally.

Hint

WACC = wE ke + wD kd(1 - tax). One-period IRR solves cost = future cash/(1 + IRR).

Work through it

1. Equity and debt weights are 0.6 and 0.4. After-tax debt cost is 6% × 0.75 = 4.5%.

2. WACC = 0.6 × 12% + 0.4 × 4.5% = 9%.

3. NPV = -100 + 109/1.09 = 0. IRR = 109/100 - 1 = 9%.

4. The project exactly earns the assumed required return: it is financially break-even at these inputs, not a strictly positive-NPV opportunity.

5. Higher leverage changes equity/debt risk and their required returns. Distress costs, tax-shield limits and financing constraints can alter value.

6. Use cash flows and discount rates consistently. Do not subtract interest again from the unlevered cash flow when discounting at WACC.

Mark your reasoning

1. 4: weights and WACC

2. 3: NPV

3. 2: IRR

4. 2: break-even decision

5. 3: leverage and consistency

Welch Corporate Finance: capital budgeting, financing and capital structure. This is a syllabus extension, not a historical midterm replica.

My mark: ____ / 14. My next correction: _____

Solution 8: Trace losses and choose a risk response

First, the words

A tranche absorbs losses according to seniority. A risk measure summarizes a distribution; it does not remove exposure.

A smaller problem first

If the equity tranche absorbs the first $5 of loss, a $3 pool loss never reaches mezzanine or senior investors.

Hint

The worst 5% includes all of the 2% at loss 40 and 3 percentage points from the loss-10 outcomes.

Work through it

1. Equity loses 5, mezzanine loses 7, senior loses 0. Remaining tranche values under this simple loss-only allocation are 0, 3 and 85.

2. Expected loss = 0.90 × 0 + 0.08 × 10 + 0.02 × 40 = $1.6m.

3. Cumulative probability is 90% at 0 and 98% at 10, so 95% VaR = $10m.

4. Worst-5% average = (0.02 × 40 + 0.03 × 10)/0.05 = $22m.

5. Stress testing can examine losses beyond an estimated distribution, correlation breakdowns, illiquidity and funding pressures. It complements a quantified limit rather than proving the model correct.

Mark your reasoning

1. 3: waterfall

2. 2: expected loss

3. 2: VaR

4. 3: ES

5. 2: stress rationale

Valuation corporate bonds/CDOs lecture; later corporate-risk-management syllabus topic. Risk course assessment remains report/presentation based.

My mark: ____ / 12. My next correction: _____

Risk Management Sketchbook

Updated November 1, 2026

A beginner-first print companion: explanations, vocabulary, formulas, worked examples, checks, and the practice relevant to this subject.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Your reading path

1. Volatility, diversification, VaR and ES
2. Forward rates, duration and convexity
3. Immunization and asset-liability management
4. Currency and commodity risk
5. Greeks and delta hedging
6. Credit risk: default, loss and spreads
7. Counterparty risk and CVA
8. Real options and the Merton model
9. Operational risk and controls
10. Bank capital and liquidity rules
11. P vs Q, risk drivers and stress tests

Then: paper labs where available, lesson-check answers, and subject practice.

Lesson 1: Volatility, diversification, VaR and ES

VaR answers "how bad is a bad day?" Expected Shortfall answers "and if it gets worse than that, how bad on average?"

Start with an everyday idea

A weather forecast can tell you a storm threshold without saying how severe the worst storms are. VaR is a loss threshold; expected shortfall looks at the average loss beyond a tail threshold.

Words you need

Volatility describes variation. Diversification spreads exposure. VaR means value at risk for a stated time and confidence. ES means expected shortfall. Neither is a maximum possible loss.

Build the explanation

Volatility (σ): the standard deviation of returns. Bigger σ = wider swings.

Diversification: two assets that don't move together have a portfolio σ below the average of their σs. Correlation ρ controls how much.

VaR at 99%: the loss you will not exceed on 99 out of 100 days. For a normal distribution it is 2.326 × σ.

Expected Shortfall (ES): the average loss on the worst 1% of days. Always bigger than VaR, and it sees how fat the tail is. Regulators now prefer ES.

Formula and units

Portfolio variance: $\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2 \cdot w_1 \cdot w_2 \cdot \rho \cdot \sigma_1 \cdot \sigma_2$

Normal VaR: $VaR = z \times \sigma \times Value$ ($z = 1.645$ at 95%, 2.326 at 99%)

Normal ES: $ES = \sigma \times Value \times \varphi(z) / (1 - confidence)$

w = weights, ρ = correlation, φ = the normal bell-curve height.

Walk through the sketch

1. This bell curve is your daily profit and loss. Most days are near zero.
2. The red sliver on the left is the worst 1% of days.
3. VaR is where that sliver starts: a loss of $465k.
4. ES is the average of everything inside the sliver: $533k. Further out than VaR.
5. Diversification: mixing two assets gives 19.35% risk, less than the 24% average.

Worked example

A $10m portfolio has a daily σ of 2%. Find the 99% 1-day VaR and ES. Then: 60/40 in two assets with σ = 20% and 30%, ρ = 0.3. Portfolio σ?

1. Daily $σ = 0.02 × 10m = $200,000

2. VaR = 2.326 × 200,000 = $465,000

3. ES = 200,000 × φ(2.326)/0.01 = 200,000 × 2.665 = $533,000

4. $\sigma_p^2 = 0.36(0.04) + 0.16(0.09) + 2(0.6)(0.4)(0.3)(0.2)(0.3) = 0.03744$, $\sigma_p = 19.35\%$, below the 24% weighted average.

VaR $465k, ES $533k; portfolio σ 19.35%

Common mistakes

1. Calling VaR the worst possible loss. It is a threshold, not a maximum.

2. Scaling 1-day VaR to 10 days by 10 instead of √10.

3. Forgetting the 2·w1·w2 in the variance formula.

Try before checking

A one-day 95% VaR of $10 means, under the model…

1. Loss can never exceed $10

2. About 5% of days have losses beyond the threshold

3. The average daily loss is $10

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6123_INFO_01_Syllabus_Fall_2026.docx (RMN1); Risk lecture decks RMN1-2 are not in your folder yet

Lesson 2: Forward rates, duration and convexity

Duration tells you how much a bond price drops when rates rise. Convexity is the correction, because the real price curve bends.

Start with an everyday idea

A fixed-rate bond is less attractive when new bonds pay more. Its price tends to fall. Duration estimates how strongly its price responds to a small rate change.

Words you need

Yield is a rate implied by price and payments. Modified duration measures price sensitivity. Convexity adjusts for curvature. A forward rate is a rate implied today for a future period.

Helpful earlier topics: Bonds and building a zero curve

Build the explanation

Forward rate: the rate for a future period that today's zero rates lock in. If 1 year pays 3% and 2 years pays 4%, the second year alone must pay 5% (continuous rates).

Duration is the weighted average time you wait for your cash, where the weights are each payment's share of the price. It is also the sensitivity: a 1% rise in yield cuts the price by about D%.

Convexity: the price-yield curve is bowed. A straight-line (duration) estimate always underestimates the price after a big move either way. Full repricing is the true answer; duration is the quick estimate.

Formula and units

Forward rate: $F(T1,T2) = (R2 \cdot T2 - R1 \cdot T1) / (T2 - T1)$

Duration: $D = \Sigma\ t \times [CF(t) \cdot e^{(-y \cdot t)}] / B$

Price change: $\Delta B \approx -B \times D \times \Delta y\ (+ \frac{1}{2} \times B \times C \times \Delta y^2)$

R = zero rates, y = bond yield (continuous), B = bond price, C = convexity.

Walk through the sketch

1. The price of the bond at each yield. It slopes down and bends (is "convex").
2. At today's yield of 12%, the price is $94.21.
3. Duration draws the tangent line at that point. Slope = $-B \times D$.
4. The tangent is below the curve on both sides. That gap is convexity.
5. For a small move the line is close. For a big move, add the convexity term or reprice fully.

Worked example

3-year bond, 10% coupon paid twice a year, yield 12% continuous (Hull's example). Price, duration, and the effect of a 0.1% rise in yield?

1. Cash flows: $5 at 0.5, 1, 1.5, 2, 2.5 years and $105 at 3 years.

2. Discount each at e^(-0.12t) and add: B = 94.213

3. Weight each time by its share of the price and add: D = 2.653 years

4. ΔB ≈ -94.213 × 2.653 × 0.001 = -0.250, so the price falls to about 93.963.

B = 94.21, D = 2.653, price drops about $0.25

Common mistakes

1. Using Δy = 0.1 instead of 0.001.

2. Forgetting the minus sign: rates up means price down.

3. Using duration for a large rate shock without the convexity term.

Try before checking

Modified duration is 5 and yield rises 0.01 (one percentage point). Approximate price change?

1. -5%

2. +5%

3. -0.05%

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx; Risk syllabus RMN2 (duration, convexity, term structure)

Lesson 3: Immunization and asset-liability management

A pension fund or bank owes money in the future. Immunization picks assets whose value moves the same way as those debts when rates change, so a rate move does not hurt.

Start with an everyday idea

If you owe money in the future, choose assets that respond to rates similarly to that obligation. This can reduce the effect of small rate changes on your ability to pay.

Words you need

A liability is an obligation. ALM means asset-liability management. Immunization matches value and rate sensitivity under assumptions; it does not remove every risk.

Helpful earlier topics: Bonds and building a zero curve, Forward rates, duration and convexity

Build the explanation

ALM (asset-liability management): manage assets and liabilities together, not separately. The risk is the gap between them.

Duration matching: make the PV of assets equal the PV of liabilities, and their durations equal. Then a small parallel rate shift changes both by about the same amount.

Convexity: also have asset convexity ≥ liability convexity, so bigger moves help rather than hurt.

Rebalance: durations drift as time passes and rates move, so immunization must be redone.

Key-rate duration: the curve does not always shift in parallel. Key-rate durations measure exposure to each part of the curve (2y, 5y, 10y…) separately.

Hedge accounting (RMN8): rules that let a company show a hedge and the item it hedges together in its accounts, so the P&L does not jump around.

Formula and units

Match PV: $PV(\text{assets}) = PV(\text{liabilities})$

Match duration: $w_1 \cdot D_1 + w_2 \cdot D_2 = D_L$, $w_1 + w_2 = 1$

Surplus change: $\Delta S \approx -(A \cdot D_A - L \cdot D_L) \times \Delta y$

w = share of money in each bond, D = duration, A and L = PV of assets and liabilities.

Walk through the sketch

1. A timeline. You owe $100m in present-value terms at year 5.

2. You can only buy bonds that pay at year 2 and year 10.

3. Think of a seesaw balanced at year 5: 62.5% at year 2 and 37.5% at year 10 balance it.

4. Now if rates move a little, assets and the liability move by about the same amount. You are immunized.

Worked example

You owe $100m (PV) in 5 years. You can buy 2-year and 10-year zero-coupon bonds. How much of each?

1. Zero-coupon bond duration = its maturity: D1 = 2, D2 = 10.

2. Solve w1 × 2 + w2 × 10 = 5 with w1 + w2 = 1.

3. w2 × 8 = 3 → w2 = 0.375, w1 = 0.625

4. Buy $62.5m of 2-year zeros and $37.5m of 10-year zeros (in PV). This "barbell" also has more convexity than the liability.

62.5% in 2-year, 37.5% in 10-year

Common mistakes

1. Matching maturities instead of durations.

2. Forgetting to also match PV.

3. Assuming the hedge lasts forever; durations drift and must be rebalanced.

Try before checking

Matching duration is intended mainly to reduce…

1. Small interest-rate-shock risk

2. Every possible loss

3. All default risk

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN2 and RMN8 (duration, immunization, ALM, hedge accounting); FRE6073_LECTURE_06_Lecture_05_Duration_and_Swaps.pptx

Lesson 4: Currency and commodity risk

Exchange-rate forwards follow interest rate parity: the currency with the higher interest rate trades at a discount forward. Commodity forwards add storage costs and a convenience yield.

Start with an everyday idea

You will pay euros next month but earn dollars. Even if the euro bill stays the same, its dollar cost can change before you pay.

Words you need

FX means foreign exchange. Exposure is the amount affected by a risk. Commodity carry includes financing and storage, offset by benefits of holding the physical good.

Helpful earlier topics: Forward price by no-arbitrage

Build the explanation

FX quote: EUR/USD = 1.10 means 1 euro costs 1.10 dollars.

Covered interest rate parity: you cannot earn free money by borrowing in one currency, converting, investing in the other, and locking in the rate back. So F = S·e(r-rf)T.

FX exposure: a US firm with euro revenue loses if the euro falls. Hedge by selling euros forward.

Commodities: holding oil costs storage (u). But having it on hand also has a benefit, the convenience yield (y). F = S·e(r+u-y)T.

Contango: F > S (storage dominates). Backwardation: F Equity risk (also RMN3): value a stock with dividends or factors (CAPM, multi-factor).

Formula and units

FX forward: F = S × e^((r - r_f)·T)

Commodity forward: F = S × e^((r + u - y)·T)

S = spot price (in domestic currency per unit), r = domestic rate, r_f = foreign rate, u = storage cost rate, y = convenience yield.

Walk through the sketch

1. Path 1: keep $1.10 in dollars at 5% for a year. You end with 1.10 × e^0.05.

2. Path 2: change it to €1, earn 3% in euros, and convert back at the forward rate F.

3. Both paths must give the same dollars, or there is free money. So F = 1.10 × e^0.02 = 1.1222.

4. Commodities: storage pushes forward prices up (contango); a convenience yield pulls them down (backwardation).

Worked example

EUR/USD spot is 1.10. USD rate 5%, EUR rate 3% (continuous). What is the 1-year forward? A US exporter will receive €1m in a year. What can it lock in?

1. F = 1.10 × e^((0.05 - 0.03) × 1) = 1.10 × e^0.02

2. e^0.02 = 1.0202 → F = 1.1222 USD per EUR

3. The euro is at a premium because its interest rate is lower.

4. Sell €1m forward: locks in $1,122,200 no matter where the euro goes.

F = 1.1222; locks in $1.1222m

Common mistakes

1. Flipping r and r_f (check: the lower-rate currency is at a forward premium).

2. Mixing up which currency the quote is "per".

3. Leaving out convenience yield when the futures curve is downward sloping.

Try before checking

You owe €100. EUR/USD rises from 1.10 to 1.20. Dollar cost…

1. Falls $10

2. Rises $10

3. Does not change

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN3 (FX exposures and parity, equity, commodity carry and basis); FRE6073_LECTURE_03_Lecture_02_Forward_and_Futures_Prices.pptx

Lesson 5: Greeks and delta hedging

A trader who sells options removes most of the risk by holding delta shares of the stock, then adjusting as delta changes. Gamma says how often you need to adjust.

Start with an everyday idea

If an option moves about half as much as its stock, half a share can offset that small price movement in a suitable opposite position. But the relationship changes, so hedges need updating.

Words you need

Delta is first-order price sensitivity. Gamma measures how delta changes. Rebalancing means adjusting the hedge. Delta-neutral means near-zero local sensitivity, not zero risk.

Helpful earlier topics: Black-Scholes and the Greeks, Hedging with futures

Build the explanation

Delta-neutral: total delta of the book = 0, so small stock moves do not change its value.

Sold calls have negative delta, so you buy shares to hedge. Sold puts have positive delta, so you sell shares.

Gamma: delta changes as the stock moves. New delta ≈ old delta + $\Gamma \times \Delta S$. High gamma = rebalance often.

A short-option, delta-hedged book has negative gamma: it loses on big moves, and earns theta (time decay) on quiet days.

Gamma/vega hedging: stock has zero gamma and vega, so you need other options to hedge those.

Hedging error: you only rebalance at discrete times and volatility is not what you assumed, so the hedge is never perfect.

Formula and units

Shares to hold = -(option position) × Δ

New $\Delta \approx \Delta + \Gamma \times \Delta S$

P&L of hedged short option $\approx \theta \cdot \Delta t - \frac{1}{2} \cdot \Gamma \cdot (\Delta S)^2$ (per option, sign for a short)

Δ = delta, Γ = gamma, θ = theta (here per unit time, positive for a short option).

Walk through the sketch

1. Delta of a call (K = 40) against the stock price. It goes from 0 to 1.

2. At S = 42 delta is about 0.78. Short 1 call → buy 0.78 shares.

3. The stock rises to 46. Delta has climbed (that climb is gamma). You must buy more shares.

4. The steepest part of the curve is near the strike: that is where gamma is highest and you rebalance most.

Worked example

You sold 1,000 calls with delta 0.60 and gamma 0.04. How many shares to buy? The stock then rises $1.25. What do you do?

1. Option delta = -1,000 × 0.60 = -600 → buy 600 shares

2. New delta ≈ 0.60 + 0.04 × 1.25 = 0.65

3. You now need 1,000 × 0.65 = 650 shares

4. Buy 50 more. (Notice: you buy after a rise and sell after a fall. That is how negative gamma costs money.)

Buy 600 shares, then 50 more

Common mistakes

1. Selling shares to hedge a short call (it is buy).

2. Thinking delta-neutral means no risk; gamma and vega remain.

3. Forgetting the contract multiplier (1 contract often = 100 shares).

Try before checking

A short call has delta -0.6 per share unit. A local hedge is…

1. Buy 0.6 share units

2. Sell 0.6 share units

3. Ignore the stock

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Hull 8e, Chapter 18 (The Greek letters); Risk syllabus RMN4 (Greeks, dynamic hedging)

Lesson 6: Credit risk: default, loss and spreads

Expected credit loss is chance of default × share you lose × amount at stake. A credit spread is the extra yield lenders demand to cover that.

Start with an everyday idea

A friend may fail to repay a loan. How often that happens, how much you would lose, and how much is owed are three separate questions.

Words you need

PD is probability of default. LGD is the fraction lost if default happens. EAD is exposure at default. Recovery is the fraction you get back.

Helpful earlier topics: Money today vs money later, Volatility, diversification, VaR and ES

Build the explanation

PD (probability of default), LGD (loss given default = 1 - recovery rate), EAD (exposure at default).

Hazard rate λ: the default "speed". Survival to time t is e-λt; PD by t is 1 - e-λt.

Credit spread ≈ λ × LGD. A CDS (credit default swap) is insurance on a bond, and its spread works out to about the same number.

P vs Q: real-world (P) default probabilities come from history. Risk-neutral (Q) ones come from prices and can differ because prices reflect risk premia and other market effects.

Expected loss is a cost you price in. Unexpected loss is what capital must absorb.

Formula and units

Survival: S(t) = e^(-λt) PD(t) = 1 - e^(-λt)
Expected loss: EL = PD × LGD × EAD
Spread: s ≈ λ × LGD → λ ≈ s / (1 - R)

λ = hazard rate per year, R = recovery rate.

Walk through the sketch

1. The chance a borrower is still alive (not defaulted) over time.
2. With λ = 2% a year, survival slides slowly down: e^(-0.02t).
3. At 5 years, 90.48% survive. So 9.52% have defaulted.
4. Expected loss stacks three things: $1m at stake × 60% lost × 1.98% chance = $11.9k.

5. The lender wants about λ × LGD = 1.2% extra yield to cover this.

Worked example

A borrower has λ = 2% a year. LGD = 60%. Exposure $1m. Find 5-year survival, 5-year PD, 1-year expected loss and the spread.

1. Survival 5y = e^(-0.02 × 5) = e^(-0.1) = 90.48%

2. PD 5y = 1 - 0.9048 = 9.52%

3. 1-year PD ≈ 1 - e^(-0.02) = 1.98%, EL ≈ 0.0198 × 0.6 × 1,000,000 = $11,900

4. Spread ≈ 0.02 × 0.6 = 1.2% = 120 basis points

Survival 90.48%, PD 9.52%, EL ≈ $11.9k, spread ≈ 120 bp

Common mistakes

1. Using recovery rate where LGD is needed (or the reverse).

2. Adding PDs across years instead of using survival.

3. Mixing P and Q probabilities in the same calculation.

Try before checking

PD 2%, LGD 50%, EAD $1,000. Expected loss?

1. $10

2. $20

3. $500

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_10_Lecture_03_Corporate_Bonds_and_CDOs.pdf; Risk syllabus RMN5-6 (PD, LGD, CDS, CVA); Hull 8e, Chapter 23 (Credit risk)

Lesson 7: Counterparty risk and CVA

If the other side of a derivative defaults while it owes you money, you lose. CVA is the price of that risk, built from exposure, default probability and loss given default.

Start with an everyday idea

A contract can be profitable on paper yet fail to pay if the other party defaults. CVA reduces its value to reflect that possibility.

Words you need

Counterparty is the other party to a contract. Exposure is what you stand to lose if they default. CVA means credit valuation adjustment. Collateral can reduce exposure.

Helpful earlier topics: Credit risk: default, loss and spreads

Build the explanation

Exposure: only positive value matters. If the trade is worth +5 to you and they default, you lose up to 5. If it is worth -5, you still owe them.

Expected exposure EE(t): the average of max(V, 0) across simulated futures. For a swap it rises then falls (a hump), because fewer payments remain later.

PFE (potential future exposure): a high percentile (e.g. 95%) of exposure. Used for limits.

CVA (credit valuation adjustment): expected loss from counterparty default, subtracted from the trade value.

Reducing it: netting, collateral (margin), and central clearing. xVA adds related adjustments (DVA, FVA, KVA).

Migration: a counterparty can be downgraded before defaulting, which raises CVA too.

Formula and units

Exposure: $E(t) = \max(V(t), 0)$

$\text{CVA} \approx \text{LGD} \times \Sigma\ \text{EE}(t_i) \times \text{PD}(t_{i-1}, t_i) \times \text{DF}(t_i)$

PD(t_{i-1}, t_i) = probability of default in that period, DF = discount factor.

Walk through the sketch

1. Possible future values of a swap. Some paths end up in your favour (above 0), some against you.

2. Only the part above zero is at risk. Averaging that gives the expected exposure: a hump over time.

3. Multiply each year's exposure by the chance of default in that year.

4. Add them up and multiply by LGD: CVA ≈ $72k.

Worked example

A 3-year swap has expected exposure $2m, $3m, $1m in years 1, 2, 3. The counterparty has a 2% chance of defaulting in each year; LGD 60%. Ignore discounting. CVA?

1. Year 1: 2 × 0.02 = 0.04

2. Year 2: 3 × 0.02 = 0.06; Year 3: 1 × 0.02 = 0.02

3. Sum = 0.12, times LGD 0.6 = 0.072

4. CVA ≈ $72,000. The trade is worth $72k less than if the counterparty were risk-free.

CVA ≈ $72,000

Common mistakes

1. Using V instead of max(V, 0).

2. Using cumulative PD for every year instead of the PD within each period.

3. Forgetting that collateral and netting cut exposure.

Try before checking

What does CVA account for?

1. The counterparty failing to pay

2. Only stock-price volatility

3. A guaranteed profit

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN5-6 (CVA, exposure profiles, xVA); Hull 8e, Chapter 24 (Credit risk)

Lesson 8: Real options and the Merton model

Choices in a business (wait, expand, abandon) are options, so they have value that plain NPV misses. Even a company's stock is an option on its assets.

Start with an everyday idea

A business can wait before building a factory, just as an option holder can wait before exercising. Flexibility can have value when the future is uncertain.

Words you need

A real option is flexibility over a business decision. Merton models a company’s equity like a call on its assets, with debt due at a future date.

Helpful earlier topics: The four derivatives and their payoffs, Credit risk: default, loss and spreads, Black-Scholes and the Greeks

Build the explanation

Real option: the right, not the duty, to take a business action later. Waiting lets you invest only if things go well.

Value it like Class 16: build a tree of the project value, find risk-neutral p, roll back.

If the option value is bigger than investing now (NPV today), wait.

Energy example (VL14): a power plant is a call on the spread between power and fuel prices: run it only when profitable.

Merton model: equity = call option on the firm's assets with strike = debt. Default happens if assets end below the debt. Distance to default counts how many σ the firm is from that line.

Formula and units

Risk-neutral p = [V0·(1 + rf) - V_down] / (V_up - V_down)

Option value = [p·max(V_up - I, 0) + (1-p)·max(V_down - I, 0)] / (1 + rf)

Merton: Equity = V·N(d1) - D·e^(-rT)·N(d2)

DD = [ln(V/D) + (μ - σ²/2)T] / (σ√T), PD = N(-DD)

V = value of project or firm assets, I = investment cost, D = face value of debt.

Walk through the sketch

1. Today the project is worth $105 and costs $100. NPV = $5.

2. Next year it is worth $140 (good) or $80 (bad).

3. If you wait, you invest only in the good case: gain $40. In the bad case you walk away: $0.

4. Rolled back with p = 0.504: the right to wait is worth $19.21. Waiting wins.

5. Same idea for a firm: shareholders get max(Assets - Debt, 0). Equity is a call on the assets.

Worked example

A project is worth $105 today and costs $100. Next year it will be worth $140 or $80; rf = 5%. Invest now, or keep the right to invest next year (cost still $100)?

1. NPV now = 105 - 100 = 5

2. p = (105 × 1.05 - 80) / (140 - 80) = 30.25/60 = 0.504

3. Wait: next year invest only if up: payoff 40 or 0

4. Option value = 0.504 × 40 / 1.05 = 19.21. Bigger than 5, so wait.

5. Merton check: V = 100, D = 80, μ = 6%, σ = 25%, T = 1 → DD = (0.2231 + 0.0288)/0.25 = 1.01, PD = N(-1.01) ≈ 15.7%

Waiting is worth $19.21 vs $5 now. Merton PD ≈ 15.7%

Common mistakes

1. Using the real-world probability in the roll-back.

2. Comparing the option with zero instead of with investing now.

3. In Merton, using the stock's volatility as the asset volatility.

Try before checking

A factory project can wait one year for more information. That flexibility…

1. May have value

2. Must be worthless

3. Guarantees profit

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE6103_LECTURE_11_Options_in_a_valuation_context.pdf; FRE6103_LECTURE_14_VL14_Real_Options_in_Energy.pdf; FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf; Risk syllabus RMN6 (Merton, distance to default)

Lesson 9: Operational risk and controls

Operational risk is loss from failed people, processes, systems or outside events: a fat-finger trade, a hack, a payment sent twice, fraud.

Start with an everyday idea

A correct trading idea can still lose money if someone enters an extra zero or a system fails. Operational risk is about failures in people, processes, systems and external events.

Words you need

A control prevents or detects mistakes. Reconciliation compares independent records. A KRI is a key risk indicator: a warning signal, not proof that a loss will happen.

Build the explanation

Basel definition: loss from inadequate or failed internal processes, people and systems, or from external events. Includes legal risk.

Frequency × severity: many small losses (errors) and rare huge ones (fraud, rogue trading). The rare ones dominate the tail.

Controls: preventive (stop it happening: approvals, access limits) and detective (catch it: reconciliations, alerts).

KRIs (key risk indicators): early-warning numbers, e.g. failed trades per day, staff turnover, overdue reconciliations.

Three lines of defense: 1) the business owns the risk, 2) risk & compliance oversee, 3) internal audit checks independently.

Payments and settlement risk: paying before you get paid (Herstatt risk). Fixed by delivery-versus-payment and CLS for FX.

Technology/cyber risk and model risk sit here too.

Formula and units

Expected annual loss ≈ expected frequency × expected severity

Residual risk = inherent risk - effect of controls

Op-risk capital looks at the tail of the loss distribution, not just the average.

Walk through the sketch

1. A bow-tie: the risk event sits in the middle (e.g. a payment sent twice).
2. On the left are the causes: people, process, systems, external events.
3. Preventive controls sit between causes and the event. They try to stop it.
4. On the right are the consequences. Detective controls and recovery plans limit the damage.

Worked example

A desk has about 12 settlement errors a year, each costing $50,000 on average. A new reconciliation control would catch 75% of them and costs $200,000 a year. Is it worth it (just on expected loss)?

1. Expected loss now: 12 × 50,000 = $600,000 a year
2. Losses avoided: 75% × 600,000 = $450,000
3. Net benefit: 450,000 - 200,000 = $250,000 a year
4. Yes, and it also cuts tail risk (fewer chances of one giant error).

Worth it: saves about $250,000 a year

Common mistakes

1. Looking only at average losses and ignoring rare severe events.
2. Confusing a control (an action) with a KRI (a measurement).
3. Treating operational risk as only IT risk.

Try before checking

Which is an operational-risk control?

1. Two-person approval for a large payment
2. Predicting stock prices
3. Buying only high-return assets

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN7 (operational risk, payments, controls, KRIs, technology risk)

Lesson 10: Bank capital and liquidity rules

Banks need capital to absorb losses and liquid assets to meet payments. The ratios below are simplified Basel-style classroom examples, not a bank-specific compliance assessment.

Start with an everyday idea

A bank needs both a cushion to absorb losses and cash-like assets to meet near-term payments. Being solvent and being able to pay today are related but different.

Words you need

Capital absorbs losses. Liquidity means meeting payments when due. RWA means risk-weighted assets. Course ratio examples are simplified; applicable rules depend on bank and jurisdiction.

Helpful earlier topics: Volatility, diversification, VaR and ES, Credit risk: default, loss and spreads

Build the explanation

Capital: equity that can absorb losses. CET1 (common equity tier 1) is the best quality.

Risk-weighted assets (RWA): each asset is weighted by riskiness (cash ~0%, mortgages lower, corporate loans ~100%). Includes market and operational risk too.

Illustrative baseline ratios: CET1 ≥ 4.5% of RWA, Tier 1 ≥ 6%, total capital ≥ 8%, plus a 2.5% conservation buffer (and extra for big banks).

Leverage ratio: Tier 1 / total exposure (not risk weighted) ≥ 3%. A backstop if risk weights are wrong.

LCR (liquidity coverage ratio): high-quality liquid assets / net cash outflows over 30 days of stress ≥ 100%.

NSFR (net stable funding ratio): stable funding / required stable funding ≥ 100% over one year.

Formula and units

CET1 ratio = CET1 / RWA (≥ 4.5%, ≥ 7% with buffer)
Leverage = Tier 1 / Total exposure (≥ 3%)
LCR = HQLA / 30-day net outflows (≥ 100%)

HQLA = high-quality liquid assets (cash, central bank reserves, government bonds).

Walk through the sketch

1. A bank balance sheet: assets on the left, how they are funded on the right.

2. Most funding is deposits and debt. A thin slice is equity: that is capital.

3. Weight the assets by risk to get RWA. Capital must be at least 7% of that (with the buffer).

4. Liquidity is a separate test: enough liquid assets to cover 30 days of stressed outflows.

Worked example

A bank has CET1 $60bn, RWA $800bn, total exposure $1,500bn, HQLA $120bn and stressed 30-day outflows of $100bn. Assume Tier 1 = CET1. Does it pass?

1. CET1 ratio = 60 / 800 = 7.5% ≥ 7% ✓

2. Leverage = 60 / 1,500 = 4.0% ≥ 3% ✓

3. LCR = 120 / 100 = 120% ≥ 100% ✓

4. It passes all three, but only 0.5% above the CET1 buffer: a $4bn loss would reach that threshold if RWA stayed unchanged; a larger loss would breach it.

7.5% CET1, 4% leverage, 120% LCR: passes

Common mistakes

1. Dividing capital by total assets for the CET1 ratio (use RWA).

2. Thinking capital is cash. Capital is how the bank is funded, not what it holds.

3. Mixing up liquidity (cash to pay today) and solvency (enough capital overall).

Try before checking

A bank has valuable long-term assets but cannot meet withdrawals today. The immediate problem is…

1. Liquidity

2. A high stock return

3. Too much profit

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMN9 (bank balance sheet, capital, leverage, liquidity); BIS Basel Framework (see "Watch online")

Lesson 11: P vs Q, risk drivers and stress tests

P is the real world (what actually tends to happen). Q is the risk-neutral world used for pricing. Risk managers measure losses in P, price in Q, and stress-test both.

Start with an everyday idea

There are two different questions: “What might really happen?” and “What price is consistent with today’s traded assets?” They need different probability tools.

Words you need

P labels real-world probabilities used for forecasting/risk. Q labels risk-neutral pricing probabilities. A stress test asks what happens under a chosen severe scenario, without needing to call it likely.

Helpful earlier topics: Volatility, diversification, VaR and ES, Binomial trees for options

Build the explanation

Q (risk-neutral): probabilities that make every asset earn the risk-free rate. Used for pricing (binomial trees, Black-Scholes, CDS). Not a forecast.

P (physical/real world): probabilities from history and judgement. Used for VaR, expected losses, capital.

Pricing weights differ from real-world probabilities. In many credit examples, implied Q default rates exceed historical rates because spreads include risk premia, but this is not a universal ordering of all outcomes.

Risk drivers: the few market factors that move a portfolio (rates, spreads, equity index, FX, vol). Map each position to them.

Aggregation: combine risks across desks using correlations; diversification lowers the total, but correlations jump in a crisis.

Ex-ante P&L / stress tests: apply a scenario (e.g. rates +200bp, equities -30%, spreads +150bp) and reprice everything to see the loss before it happens.

Formula and units

Q default intensity from spread: $\lambda Q \approx s / LGD$

Stress loss $\approx \Sigma$ (sensitivity to driver $\times$ shock) (or full revaluation)

Bond: $\Delta V \approx -V \times D \times \Delta y$

Use P for "how likely / how much could we lose", Q for "what is it worth".

Walk through the sketch

1. P: the real-world spread of outcomes for a risky asset. Its average is above the risk-free rate.

2. Q: the same outcomes reweighted so the average is the risk-free rate. More weight on bad outcomes.

3. Use P for risk (VaR, expected loss). Use Q for prices.

4. Stress test: shock each driver and add the hits. Bonds -$6m, equities -$9m, total -$15m.

Worked example

A bond portfolio worth $50m (duration 6) and an equity portfolio worth $30m. Stress: rates +2%, equities -30%. Loss? Also: a spread of 150bp with 60% LGD implies what Q default rate, if history says 1%?

1. Bonds: -50 × 6 × 0.02 = -$6m

2. Equities: -30% × 30 = -$9m

3. Total stress loss ≈ $15m (19% of $80m)

4. λQ ≈ 0.015 / 0.6 = 2.5% a year vs 1% in P. The gap is the credit risk premium.

Stress loss ≈ $15m; Q default rate 2.5% vs P 1%

Common mistakes

1. Using Q probabilities to forecast real default rates.

2. Adding stress losses that cannot happen together, or ignoring ones that do.

3. Trusting normal-times correlations in a stress scenario.

Try before checking

Use risk-neutral probabilities mainly to…

1. Predict the actual frequency of every event

2. Price claims consistently under a model

3. Guarantee no losses

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Risk syllabus RMNA (P vs Q, drivers, aggregation, ex-ante P&L, stress); Valuation notes: instructor-linked explanation of risk neutrality

Lesson-check answers

1. Volatility, diversification, VaR and ES

Answer: About 5% of days have losses beyond the threshold. 95% of modeled outcomes are at or below the loss threshold. The remaining tail can contain much larger losses.

2. Forward rates, duration and convexity

Answer: -5%. Approximate proportional change = -5 × 0.01 = -0.05, or -5%. This is an estimate, not exact repricing.

3. Immunization and asset-liability management

Answer: Small interest-rate-shock risk. It targets rate sensitivity. Large or non-parallel curve moves, defaults and changing cash flows can still hurt.

4. Currency and commodity risk

Answer: Rises $10. Old cost = 100 × 1.10 = $110. New cost = 100 × 1.20 = $120.

5. Greeks and delta hedging

Answer: Buy 0.6 share units. The stock contributes +0.6 delta, offsetting -0.6. Contract multipliers determine the actual share count.

6. Credit risk: default, loss and spreads

Answer: $10. 0.02 × 0.50 × 1,000 = $10. This average is not the loss in every outcome.

7. Counterparty risk and CVA

Answer: The counterparty failing to pay. CVA reflects expected loss from counterparty default, combining exposure, default probabilities and loss severity.

8. Real options and the Merton model

Answer: May have value. Waiting can avoid a bad investment, though delay can also lose revenue. The value depends on assumptions.

9. Operational risk and controls

Answer: Two-person approval for a large payment. Independent approval can catch errors or unauthorized payments before money moves.

10. Bank capital and liquidity rules

Answer: Liquidity. It may have enough asset value overall but insufficient accessible cash. Capital and liquidity solve different problems.

11. P vs Q, risk drivers and stress tests

Answer: Price claims consistently under a model. Q is a pricing device. Real-world frequencies and risk forecasts use P; neither eliminates uncertainty.

Prepare a risk report, not a guessed final exam

The supplied FRE 6123 syllabus assigns 40% to team assignments and 30% each to two individual report/presentation assignments. The RMN lecture decks were absent from the supplied pack; these notes are foundations, not substitutes for those decks.

A report rehearsal

1. Choose one clearly bounded risk exposure and decision-maker.

2. State the data source, period, units and key assumptions.

3. Explain one suitable risk measure and compute a small transparent example.

4. Stress an assumption, discuss limitations and propose a response.

5. Prepare a short presentation where every chart answers a decision question. Follow the instructor’s actual assignment brief.

Bloomberg Sketchbook

Updated November 1, 2026

A beginner-first print companion: explanations, vocabulary, formulas, worked examples, checks, and the practice relevant to this subject.

How to use this edition

Read the everyday idea first. Define the words. Work through the example with your calculator. Attempt the check or question on paper before reading its explanation. Then change an input and explain what changes.

The web app adds narration, sliders, saved answers and timers. This edition includes paper experiments with predictions, calculations and explanations. It does not contain your personal browser notes.

Original study aid based on the supplied course materials. Final scope and assessment instructions may change. Risk uses reports/presentations in the supplied syllabus; Bloomberg certification is completed in its official portal.

Your reading path

1. Bloomberg: passing the four tests
2. Economic indicators
3. Currencies
4. Fixed income and the yield curve
5. Equities and valuation multiples

Then: paper labs where available, lesson-check answers, and subject practice.

Lesson 1: Bloomberg: passing the four tests

The Bloomberg course is pass/fail on four tests: at least 70% on each one, done by December 14, under class code FM3NH456BD.

Start with an everyday idea

Bloomberg is a certification task: learn the required modules, take the official tests on Bloomberg, and verify your recorded results. This app only helps you prepare.

Words you need

BMC means Bloomberg Market Concepts. Your supplied requirements say at least 70% on each of four required tests. An app practice answer is not an official test result.

Build the explanation

Sign up with the access guide in your folder and enter class code FM3NH456BD so the instructor can see your scores.

Watching videos or getting a certificate is not enough. All four test scores must be 70% or more.

Do the lessons actively: pause, note unfamiliar terms, and link them to what you learned here (yields and duration, futures, options, credit).

Aim to finish by your own target of December 6, leaving a week of buffer before the real deadline of December 14.

After each test, check the score shows on your class record.

Formula and units

Pass = all four tests ≥ 70%

Deadline: Mon Dec 14, 2026 (target Dec 6)

Bloomberg lessons and tests are online, not in your folder.

Walk through the sketch

1. Step 1: sign up and enter class code FM3NH456BD.
2. Then four tests along the way. Each needs at least 70%.
3. Your own target finish line is December 6.
4. The real deadline is December 14. The week in between is your safety buffer.

Worked example

Plan: you have 30-minute Bloomberg slots twice a week from Nov 2. How do you fit four tests before Dec 6?

1. Week of Nov 2: register, confirm the class code, start the first module.
2. Weeks of Nov 9 and Nov 16: finish modules and take tests 1 and 2.
3. Weeks of Nov 23 and Nov 30: tests 3 and 4 (book a longer session if a test needs it).
4. Retake any test under 70%, then check all four scores appear.

Four tests done by Dec 6, one week of buffer

Common mistakes

1. Not entering the class code, so the instructor never sees your scores.
2. Assuming a certificate means you passed.
3. Leaving all four tests to the final week.

Try before checking

One test is 65% but your average is 80%. Have you met the supplied requirement?

1. Yes, the average is enough
2. No, each required test must reach 70%
3. Yes, if the videos are complete

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE5500_NOTES_01_Bloomberg_Signup_and_BMC_Access_Summer_2025.pdf; FRE5500_NOTES_02_Requirements.md

Lesson 2: Economic indicators

Markets move on surprises, not on the news itself: what matters is the actual number minus what economists expected.

Start with an everyday idea

If everyone expected 180,000 new jobs but the release says 250,000, the surprise is +70,000. Markets react to surprises and details, not just whether a headline sounds good.

Words you need

GDP measures economic output. Inflation is price growth. Consensus is the expected figure. An indicator is a measured clue about the economy.

Build the explanation

GDP: total output. Growth rate is the headline. Released quarterly.

Inflation (CPI, PCE): how fast prices rise. Central banks target about 2%.

Jobs (nonfarm payrolls, unemployment rate): monthly; one of the biggest market movers in the US.

Surveys (PMI, consumer confidence): early signals. PMI above 50 = expanding.

Leading vs lagging: leading indicators (PMI, building permits) turn before the economy; lagging ones (unemployment) after.

Central bank link: strong growth or high inflation → expect higher policy rates → bond yields up, currency up, often stocks down.

On the terminal, the economic calendar shows the release time, the consensus (survey) and the actual.

Formula and units

Surprise = Actual - Consensus

Positive surprise (strong data) → yields ↑, currency ↑

Negative surprise → yields ↓, currency ↓

Typical reactions. Real markets can react differently if the details disagree with the headline.

Walk through the sketch

1. The economists' consensus forecast: +180k jobs.

2. The release: +250k. A +70k surprise.

3. The market reacts within seconds: 2-year yield jumps up.

4. And the dollar rises against other currencies.

Worked example

Consensus for US payrolls is +180k. The actual is +250k. Unemployment is unchanged. What do you expect for Treasury yields and the dollar?

1. Surprise = 250k - 180k = +70k, a strong beat

2. Stronger jobs → the Fed is less likely to cut rates

3. Short-term Treasury yields rise (prices fall)

4. The dollar strengthens, as higher US rates attract money

Yields up, dollar up

Common mistakes

1. Reacting to the level instead of the surprise.

2. Mixing up month-over-month and year-over-year inflation.

3. Forgetting that a bond price falls when its yield rises.

Try before checking

Expected jobs 180k, actual 250k. Surprise?

1. +70k

2. +250k

3. -70k

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: FRE5500_NOTES_02_Requirements.md; Bloomberg Market Concepts: Economic Indicators module

Lesson 3: Currencies

An exchange rate is just the price of one currency in another. Interest rates, inflation, growth and risk appetite push it around.

Start with an everyday idea

A currency quote is a price tag with two currency units. EUR/USD 1.10 says one euro costs 1.10 US dollars. Write the units before multiplying.

Words you need

Base currency is the first currency in the pair. Quote currency is the second. Appreciation means gaining value relative to the other currency.

Build the explanation

Quote: EUR/USD = 1.10 means 1 EUR (the base) = 1.10 USD (the quote currency). If it rises to 1.12, the euro got stronger.

Pips: the 4th decimal (0.0001). Spread = ask - bid.

Interest rate differentials: higher rates attract money → currency tends to strengthen (spot). Forwards follow interest rate parity (see the Risk class on currencies).

Carry trade: borrow in a low-rate currency, invest in a high-rate one. Earns the gap, until a sudden reversal.

PPP (purchasing power parity): in the long run, the same basket should cost the same everywhere; high-inflation currencies weaken.

Safe havens (USD, JPY, CHF) tend to rise in a crisis.

Who cares: exporters, importers, investors holding foreign assets, central banks.

Formula and units

Amount in quote currency = amount in base × rate

Cross rate: EUR/JPY = EUR/USD × USD/JPY

% change of base = new rate / old rate - 1

Always write the pair as BASE/QUOTE and keep units: QUOTE per 1 BASE.

Walk through the sketch

1. A seesaw: EUR on the left, USD on the right. At 1.10, one euro balances 1.10 dollars.

2. The ECB raises rates; money flows into euros. The euro side gets heavier: EUR/USD rises to 1.155.

3. Cross rates: if you know EUR/USD and USD/JPY you can get EUR/JPY by multiplying.

Worked example

EUR/USD = 1.10 and USD/JPY = 150. (a) How many dollars is €2m? (b) What is EUR/JPY? (c) EUR/USD moves to 1.155. How much did the euro gain?

1. (a) 2,000,000 × 1.10 = $2,200,000

2. (b) EUR/JPY = 1.10 × 150 = 165 yen per euro

3. (c) 1.155 / 1.10 - 1 = +5%

4. So a US investor holding euro assets gained 5% from the currency alone.

$2.2m; EUR/JPY 165; euro +5%

Common mistakes

1. Multiplying when you should divide (check units: USD per EUR × EUR = USD).

2. Reading a fall in EUR/USD as the dollar weakening (it is the dollar strengthening).

3. Forgetting the currency effect on foreign investments.

Try before checking

At EUR/USD 1.10, €20 buys how many dollars?

1. $18.18

2. $22

3. $20

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Bloomberg Market Concepts: Currencies module

Lesson 4: Fixed income and the yield curve

A bond's price and yield move in opposite directions. The yield curve (yields by maturity) shows what the market expects for rates and growth.

Start with an everyday idea

A yield curve compares borrowing rates for different lengths of time. A credit spread compares a riskier borrower’s yield with a suitable reference yield.

Words you need

Maturity is time until repayment. A basis point is 0.01 percentage point: 100 bp = 1 percentage point. Inverted means shorter-term yields exceed longer-term yields.

Helpful earlier topics: Bonds and building a zero curve, Forward rates, duration and convexity

Build the explanation

Price vs yield: rates up → existing bonds with lower coupons are worth less → price down.

Yield curve shapes: normal (upward: longer = higher yield), flat, inverted (short rates above long, often seen before recessions).

Central banks set short rates; long rates reflect expected future short rates plus a term premium.

Credit spread: corporate bond yield - Treasury yield of the same maturity. Wider spread = more perceived default risk.

Ratings: AAA ... BBB- is investment grade; BB+ and below is high yield ("junk").

Duration (see the Derivatives class) tells you price sensitivity: duration 7 → about -7% for +1% in yield.

Formula and units

Credit spread = corporate yield - Treasury yield
Curve slope = 10-yr yield - 2-yr yield
% price change ≈ -Duration × Δyield

1 basis point (bp) = 0.01%.

Walk through the sketch

1. A normal yield curve: you get paid more to lend for longer.

2. A flat curve: all maturities pay about the same.

3. An inverted curve: short rates above long rates. Markets expect cuts, often a recession warning.

4. Add a company's curve above the Treasury curve. The gap is the credit spread: 130 bp at 10 years.

Worked example

10-year Treasury yields 4.20%. A BBB company's 10-year bond yields 5.50%. The 2-year Treasury yields 4.60%. Spread? Curve shape? If the corporate bond (duration 8) sees its yield rise 0.25%, price change?

1. Credit spread = 5.50 - 4.20 = 1.30% = 130 bp

2. Slope = 4.20 - 4.60 = -40 bp → the curve is inverted

3. Price change ≈ -8 × 0.25% = -2%

4. Inverted curve: markets expect rate cuts ahead, often linked with slowing growth.

Spread 130 bp; inverted curve; price -2%

Common mistakes

1. Saying a bond price rises when yields rise.

2. Comparing yields of different maturities when computing a spread.

3. Writing 130 bp as 13%.

Try before checking

Corporate yield 5.5%, Treasury yield 4.2%. Spread?

1. 13 bp

2. 130 bp

3. 1,300 bp

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Bloomberg Market Concepts: Fixed Income module; FRE6103_LECTURE_04_Lecture_2B_Bonds_and_rates.pdf

Lesson 5: Equities and valuation multiples

A share is a slice of a company's future profits. Multiples like P/E are a quick way to compare how "expensive" companies are.

Start with an everyday idea

A share is a small ownership stake. If a share costs $50 and annual earnings per share are $2.50, you pay 20 times that annual earnings amount. That is a comparison, not a guaranteed payback time.

Words you need

EPS is earnings per share. P/E is price divided by EPS. A dividend is a cash distribution. Enterprise value includes debt and subtracts cash in the simplified formula.

Helpful earlier topics: Money today vs money later, Portfolio theory and the CAPM

Build the explanation

P/E = price / earnings per share. How many dollars you pay for $1 of yearly profit. Higher can mean faster expected growth, or overpricing.

EV/EBITDA: enterprise value (equity + debt - cash) over operating earnings. Compares firms with different debt levels.

Dividend yield = dividend / price. Market cap = price × shares.

DCF (discounted cash flow): the "proper" value: PV of future free cash flows (Valuation classes). A simple version is the Gordon model: P = D1/(r - g).

Indices (S&P 500, etc.) track groups of stocks. Beta measures how much a stock moves with the index (Portfolio/CAPM class).

What moves stocks: earnings surprises, guidance, interest rates (higher rates = lower PV of future profits), sector news.

Formula and units

P/E = Price / EPS

EV = Market cap + Debt - Cash

Gordon growth: P0 = D1 / (r - g)

EPS = earnings per share, D1 = next year's dividend, r = required return, g = growth rate.

Walk through the sketch

1. A $50 share. Each year it earns $2.50 per share.
2. So you pay 20 years of today's profit: P/E = 20×.
3. The Gordon model values the growing stream of dividends: 1.50 / (9% - 6%) = $50.
4. Small changes in growth move the value a lot: at 5% growth the value drops to $37.50.

Worked example

A stock is $50 with EPS $2.50 and next year's dividend $1.50. Required return 9%, dividends grow 6% a year. P/E? Gordon value? Is it cheap?

1. P/E = 50 / 2.50 = 20×
2. Gordon value = 1.50 / (0.09 - 0.06) = 1.50 / 0.03 = $50
3. Model value = market price, so fairly priced under these assumptions.
4. If growth were only 5%: 1.50 / 0.04 = $37.50, so the price assumes 6% growth.

P/E 20×; Gordon value $50 (fairly priced)

Common mistakes

1. Comparing P/E across very different industries.
2. Using Gordon when g ≥ r (the formula breaks).
3. Forgetting to subtract cash and add debt for EV.

Try before checking

Price $50, EPS $2.50. P/E?

1. 20×
2. 2×
3. 125×

Write your reasoning: _____

The answer and explanation are in the lesson-check answers section.

Reading trail: Bloomberg Market Concepts: Equities module; FRE6103_BOOK_02_Corporate_Finance_Ivo_Welch_4e_Early_Preview_2016.pdf

Lesson-check answers

1. Bloomberg: passing the four tests

Answer: No, each required test must reach 70%. The supplied October 30 requirements specify at least 70% on each of four tests, not a 70% average.

2. Economic indicators

Answer: +70k. Actual minus expected = 250k - 180k = +70k. The market response is not guaranteed.

3. Currencies

Answer: $22. 20 EUR × 1.10 USD per EUR = 22 USD. The EUR units cancel.

4. Fixed income and the yield curve

Answer: 130 bp. 5.5% - 4.2% = 1.3 percentage points = 130 basis points.

5. Equities and valuation multiples

Answer: 20×. 50 ÷ 2.50 = 20. A multiple by itself does not tell you whether a stock is a good investment.

Bloomberg: original warm-up practice

These 32 questions and four cases are original learning aids. They were not copied from real BMC tests. No official BMC assessment questions were supplied. This practice neither counts toward certification nor predicts your official score.

Complete the assigned modules and assessments in the Bloomberg for Education portal. Follow the requirements supplied for your NYU class, and verify any changes with your instructor.

Economic Indicators - try first

1. What does GDP (gross domestic product) measure?

A. The total value of goods and services a country produces in a period B. The total value of a country’s stock market C. How much the government spends each year

2. Economists expected 200,000 new jobs. The report shows 150,000. What is the surprise?

A. +150,000 B. -50,000 C. +50,000

3. What does the CPI (consumer price index) track?

A. Company profits B. Changes in the price of a typical basket of things people buy C. Interest rates on government bonds

4. What is the difference between real and nominal GDP?

A. Real GDP removes the effect of price changes (inflation) B. Real GDP only counts physical goods, not services C. There is no difference

5. Inflation is running well above the central bank’s target. What is the central bank most likely to do?

A. Cut interest rates B. Raise interest rates C. Print a new currency

6. A manufacturing PMI reading of 55 usually suggests…

A. Manufacturing is shrinking B. Manufacturing is growing C. Prices are falling

7. Which indicator is usually described as “lagging” (it changes after the economy has already turned)?

A. The unemployment rate B. Stock market prices C. New building permits

8. Why can a “good” economic number make a market fall?

A. Markets only look at bad news B. The number was still worse than what traders expected, or it raises the chance of higher interest rates C. Economic data never affects markets

Economic Indicators - explanations

1. The total value of goods and services a country produces in a period. GDP adds up the value of everything produced in the economy over a quarter or a year. It is the broadest measure of economic output.

2. -50,000. Surprise = actual - expected = 150,000 - 200,000 = -50,000. Markets react to the surprise, not just the headline number.

3. Changes in the price of a typical basket of things people buy. CPI follows the cost of a fixed basket of consumer goods and services. Its yearly percentage change is a common measure of inflation.

4. Real GDP removes the effect of price changes (inflation). Nominal GDP uses today’s prices. Real GDP adjusts for inflation, so it shows whether the economy actually produced more, not just charged more.

5. Raise interest rates. Higher policy rates make borrowing more expensive, which slows spending and helps bring inflation down.

6. Manufacturing is growing. A purchasing managers’ index above 50 signals expansion; below 50 signals contraction.

7. The unemployment rate. Firms usually hire and fire after demand has already changed, so unemployment tends to confirm a turn rather than predict it.

8. The number was still worse than what traders expected, or it raises the chance of higher interest rates. Prices already reflect expectations. A strong number can disappoint if expectations were higher, or worry investors about rate hikes.

Currencies - try first

1. EUR/USD = 1.10 means...

A. 1 US dollar buys 1.10 euros B. 1 euro costs 1.10 US dollars C. The euro is worth 10% less than the dollar

2. EUR/USD moves from 1.10 to 1.15. What happened?

A. The euro strengthened against the dollar B. The euro weakened against the dollar C. Nothing; quotes do not show strength

3. USD/JPY = 150. How many yen do you get for $1,000?

A. 6.67 yen B. 150 yen C. 150,000 yen

4. All else equal, if a country raises its interest rates, its currency tends to...

A. Strengthen, because investors want the higher return B. Weaken, because borrowing costs rise C. Stay exactly the same

5. The home currency weakens. Who usually benefits most?

A. People importing foreign goods B. Exporters selling abroad C. Tourists travelling overseas

6. Country A has a higher interest rate than country B. In the forward market, A’s currency usually trades...

A. At a forward discount (cheaper for future delivery) B. At a forward premium (more expensive for future delivery) C. At exactly the spot rate

7. During a market panic, which currencies have often been treated as “safe havens”?

A. Japanese yen and Swiss franc B. Currencies of fast-growing emerging markets C. None; all currencies fall together

8. A country imports much more than it exports for many years. All else equal, this tends to put what pressure on its currency?

A. Upward pressure B. Downward pressure C. No pressure

Currencies - explanations

1. 1 euro costs 1.10 US dollars. The first currency (EUR) is the base. The quote tells you how many units of the second currency (USD) one euro costs.

2. The euro strengthened against the dollar. One euro now buys more dollars (1.15 instead of 1.10), so the euro gained value relative to the dollar.

3. 150,000 yen. Each dollar buys 150 yen, so $1,000 × 150 = 150,000 yen.

4. Strengthen, because investors want the higher return. Higher rates attract foreign money looking for better returns. Buying that currency pushes its value up.

5. Exporters selling abroad. A weaker home currency makes exported goods cheaper for foreign buyers, which helps exporters sell more.

6. At a forward discount (cheaper for future delivery). Covered interest parity: the extra interest you earn in A is offset by buying A’s currency back more cheaply in the forward contract.

7. Japanese yen and Swiss franc. Investors under stress have historically moved money into currencies such as the yen, the Swiss franc and often the US dollar.

8. Downward pressure. To pay for imports, the country sells its own currency for foreign currency. Persistent net selling tends to weaken it over time, though capital flows can offset this.

Fixed Income - try first

1. Market interest rates (yields) rise. What happens to the price of an existing fixed-coupon bond?

A. It rises B. It falls C. It stays the same

2. A bond pays a 5% coupon. Similar bonds now yield 6%. The bond will trade…

A. Below face value (at a discount) B. At face value (par) C. Above face value (at a premium)

3. How much is one basis point?

A. 1% B. 0.1% C. 0.01%

4. The 2-year yield is above the 10-year yield. This yield curve is called…

A. Normal (upward sloping) B. Inverted C. Flat by definition

5. A company’s credit spread widens from 1.5% to 3%. Investors are signalling…

A. They see more default risk B. The company has become safer C. Government rates have fallen

6. Which bond is usually more sensitive to a change in interest rates?

A. A 2-year bond B. A 30-year bond C. Both are equally sensitive

7. A bond has a modified duration of 7. Yields rise by 1 percentage point. Roughly how does its price change?

A. Falls about 7% B. Rises about 7% C. Falls about 0.7%

8. On the S&P scale, what is the lowest rating still called “investment grade”?

A. A B. BBB- C. BB+

Fixed Income - explanations

1. It falls. New bonds now pay more, so an old bond with a lower coupon must become cheaper to compete. Bond prices and yields move in opposite directions.

2. Below face value (at a discount). Buyers will only accept the lower 5% coupon if they pay less than face value, so the price falls below par.

3. 0.01%. A basis point is one hundredth of a percentage point: 100 bp = 1%.

4. Inverted. An inverted curve means short-term rates are higher than long-term rates. Markets often read it as a warning of a future slowdown.

5. They see more default risk. The spread is the extra yield over a safe government bond. A wider spread means investors want more compensation for credit risk.

6. A 30-year bond. Longer bonds have higher duration: more of their cash arrives far in the future, so discounting changes affect them more.

7. Falls about 7%. Price change ≈ -duration × change in yield = -7 × 1% = -7%. Convexity makes the true fall a little smaller.

8. BBB-. BBB- and above is investment grade. BB+ and below is high-yield, sometimes called “junk”.

Equities - try first

1. A share costs $50 and earnings per share are $2.50. What is the P/E ratio?

A. 2.5 B. 20 C. 125

2. How is market capitalisation calculated?

A. Share price × number of shares outstanding B. Revenue × profit margin C. Total debt + cash

3. A $50 share pays $2 of dividends a year. What is the dividend yield?

A. 2% B. 4% C. 25%

4. A stock has a beta of 1.5. If the market rises 10%, the stock is expected to…

A. Rise about 15% B. Rise about 10% C. Fall about 15%

5. A company buys back some of its own shares. If total earnings stay the same, earnings per share…

A. Fall B. Rise C. Stay the same

6. Compared with “value” stocks, “growth” stocks usually have…

A. Lower P/E ratios B. Higher P/E ratios C. No earnings at all

7. A company beats this quarter’s earnings forecast, but the stock falls. A likely reason is…

A. It lowered its guidance for future quarters B. Stock prices ignore earnings C. The beat was too large

8. Market cap $100m, debt $30m, cash $10m. What is enterprise value (EV)?

A. $120m B. $140m C. $80m

Equities - explanations

1. 20. P/E = price ÷ earnings per share = 50 ÷ 2.50 = 20. Investors pay $20 for every $1 of yearly earnings.

2. Share price × number of shares outstanding. Market cap is what the market says all the company’s shares are worth together.

3. 4%. Dividend yield = annual dividend ÷ price = 2 ÷ 50 = 4%.

4. Rise about 15%. Beta measures sensitivity to the market: 1.5 × 10% = about 15%. It is an average tendency, not a guarantee.

5. Rise. The same earnings are now divided among fewer shares, so EPS rises.

6. Higher P/E ratios. Investors pay more today for growth stocks because they expect earnings to rise quickly in the future.

7. It lowered its guidance for future quarters. Prices look forward. Weak guidance about future earnings can outweigh a good quarter that has already happened.

8. $120m. EV = market cap + debt - cash = 100 + 30 - 10 = $120m. It is roughly what it costs to buy the whole business.

Market case: Read the surprise, not just the headline

Markets expected annual inflation of 3.0%. The release is 3.4%, down from 3.6% previously. Two-year government yields rise after the release.

1. Calculate the change from the previous release and the surprise versus expectations.

2. Explain how falling inflation and rising yields can occur together. What would you check before claiming causation?

Work through the explanation

1. Inflation fell by 0.2 percentage points from the previous release, but was 0.4 percentage points above expectations. Those are different comparisons.

2. Investors may revise expected policy rates upward relative to their prior expectations, even while inflation is declining. Prices respond to new information relative to what was already priced in.

3. Check release timing, core inflation, revisions, other simultaneous news and changes in expected policy. A same-day move alone does not prove which news caused it.

Market case: Keep the currency quote straight

A US student must pay EUR 2,000. EUR/USD rises from 1.08 to 1.12 USD per EUR. Ignore fees and bid/ask spreads.

1. Which currency strengthened? Calculate the dollar cost before and after.

2. What is the percentage rise in EUR/USD? Is the percentage change in its reciprocal exactly the negative of that number?

Work through the explanation

1. The euro strengthened: each euro now buys more dollars. The payment costs $2,160 before and $2,240 after, an increase of $80.

2. EUR/USD rose by 1.12/1.08 - 1 = 3.7037%.

3. The reciprocal USD/EUR changed by 1.08/1.12 - 1 = -3.5714%. Reciprocal percentage returns have different denominators, so they are not exact opposites.

Market case: Separate rate risk from credit risk

A corporate bond yields 6.0%, while a comparable government bond yields 4.0%. Later these are 6.4% and 4.6%. The corporate bond has modified duration 5.0.

1. Calculate the credit spread at each date. Did it widen or narrow?

2. Estimate the corporate bond’s percentage price change from its yield change. Explain why its price can fall even as its credit spread narrows.

Work through the explanation

1. Initial spread = 2.0 percentage points (200 bp); later = 1.8 points (180 bp). It narrowed by 20 bp.

2. The corporate yield rose 0.4 percentage points. Approximate price change = -5 × 0.004 = -2.0%, ignoring convexity and coupon income.

3. The government yield rose 60 bp while the spread narrowed 20 bp, leaving a net corporate-yield rise of 40 bp. Narrower spreads do not guarantee rising prices.

Market case: Read an earnings headline critically

A company earns $100m and has 50m shares. Its stock is $40. It buys back 5m shares; assume earnings and stock price stay unchanged for this exercise.

1. Calculate EPS and P/E before and after the buyback.

2. Has the business necessarily created economic value just because EPS rose? Name information you would need.

Work through the explanation

1. Before: EPS = 100/50 = $2; P/E = 40/2 = 20.

2. After: 45m shares remain, EPS = 100/45 = $2.2222 and P/E = 40/2.2222 = 18.

3. EPS increased because the denominator shrank. Examine the repurchase price, financing, lost interest income, investment alternatives and leverage. Holding earnings and price fixed isolates the arithmetic, not the full economic effect.